

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

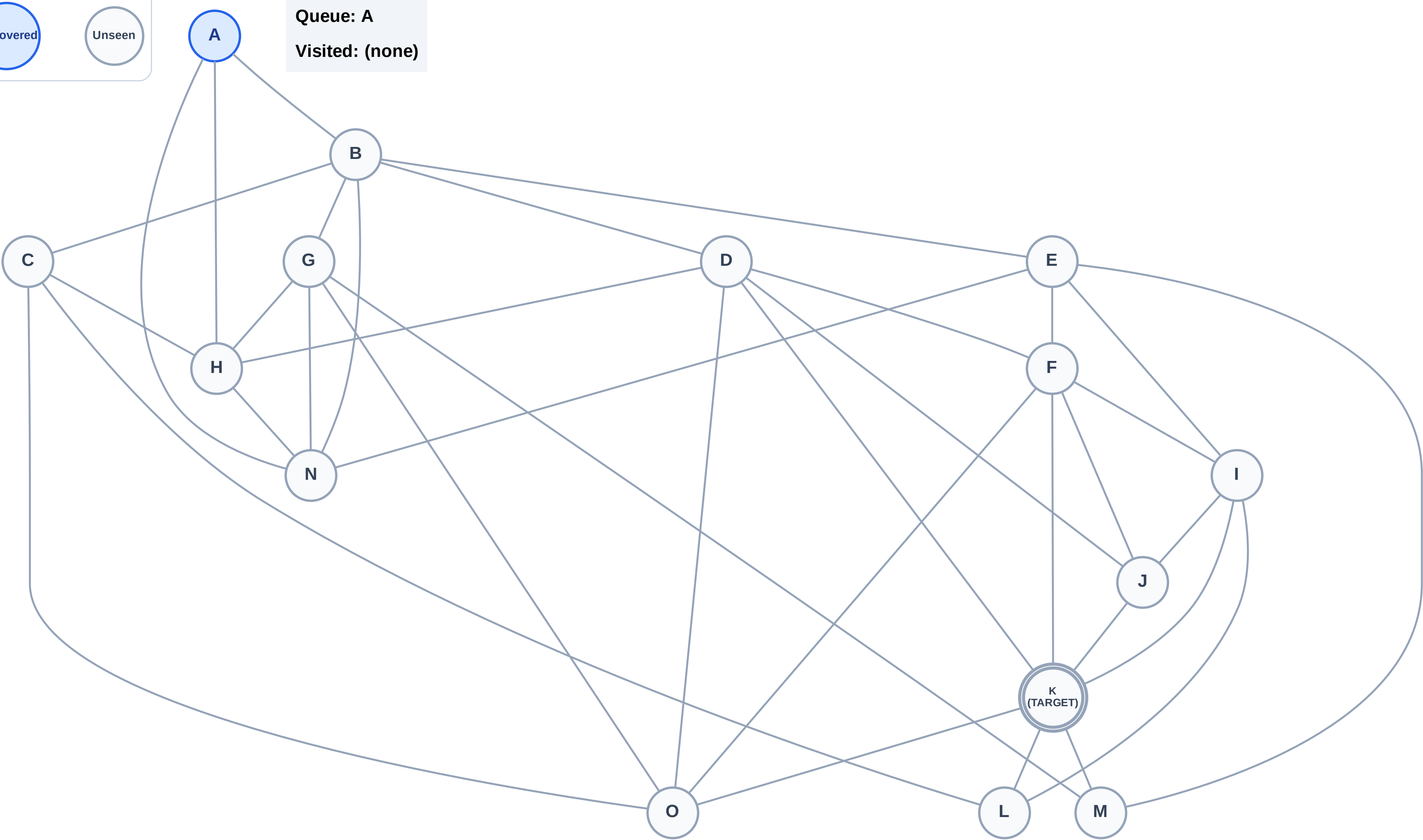
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

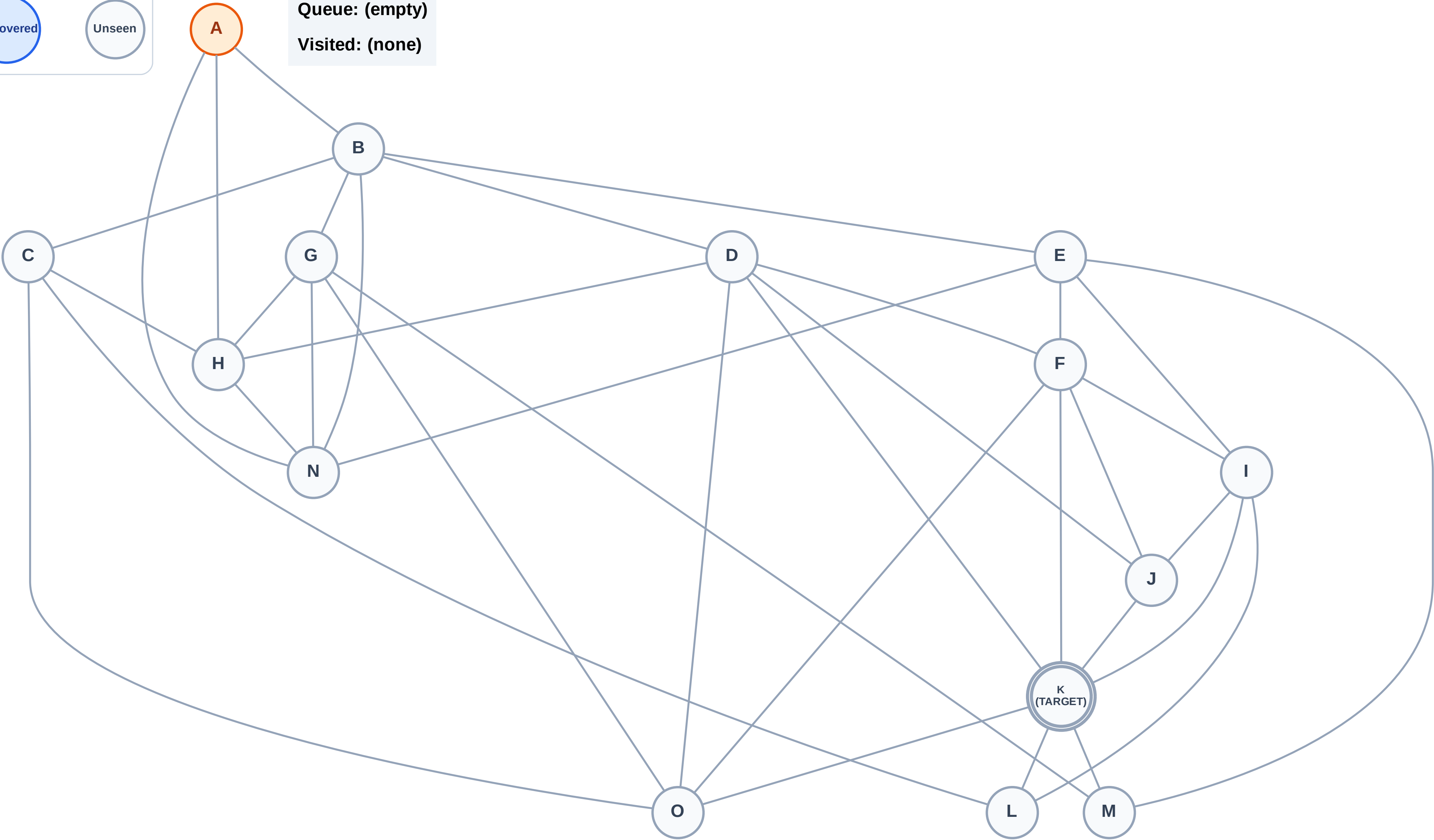
Complete

Processing

Discovered

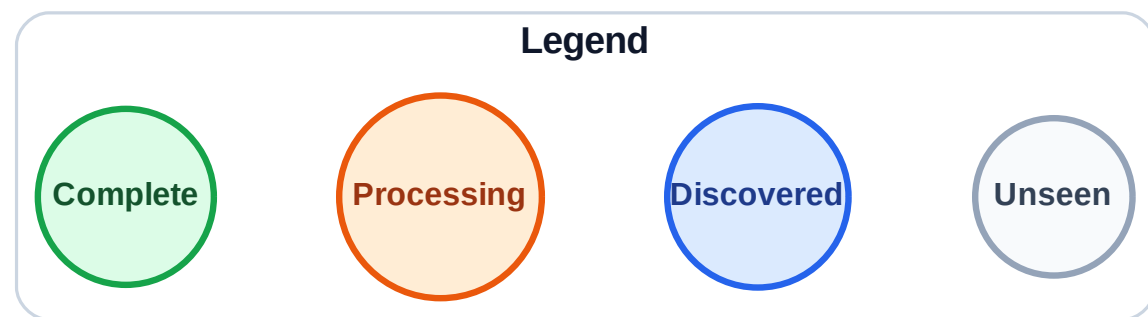
Unseen

Queue: (empty)
Visited: (none)

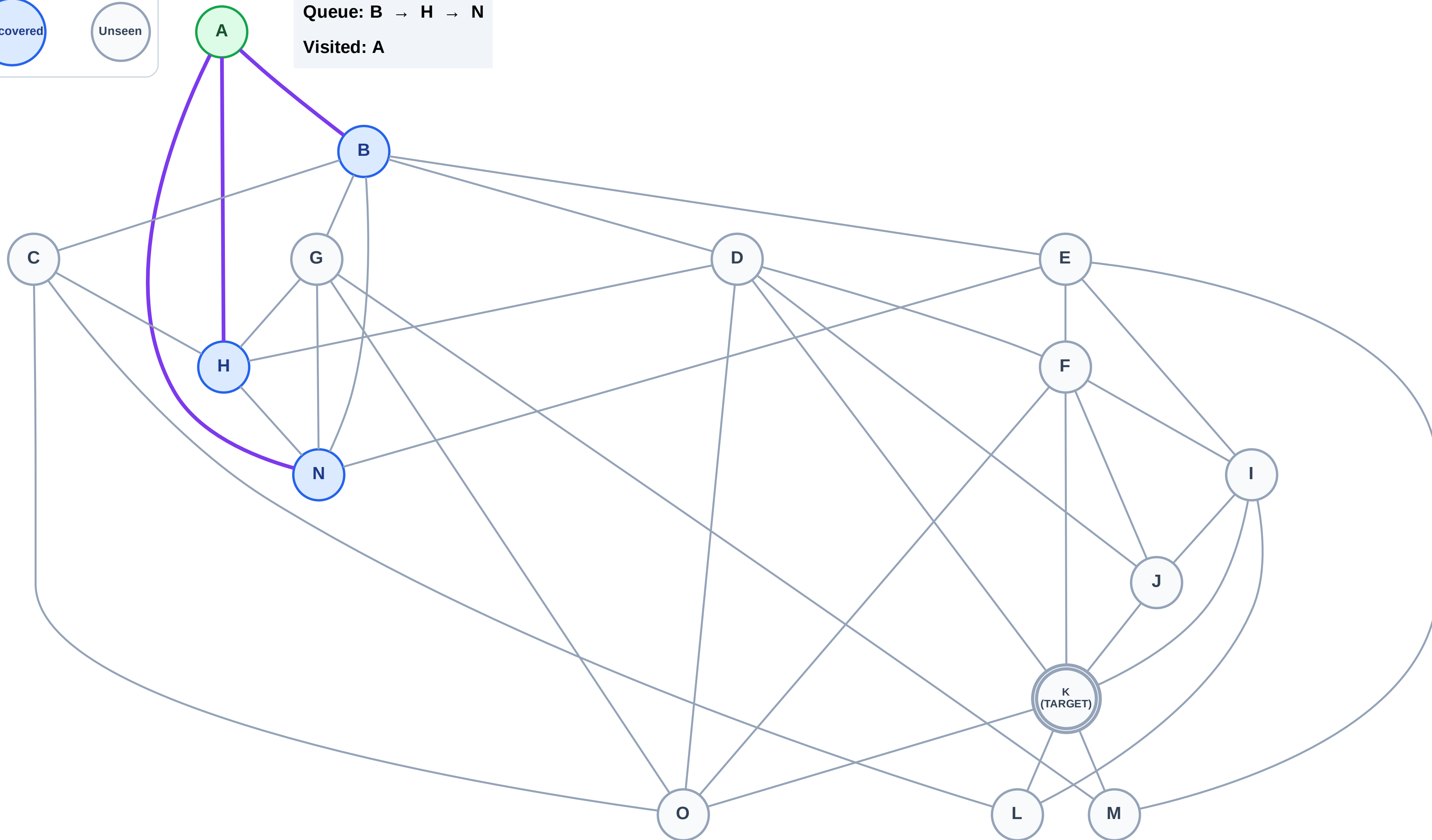


BFS Traversal

Step 3: Discover B, H, N from A



Queue: B → H → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

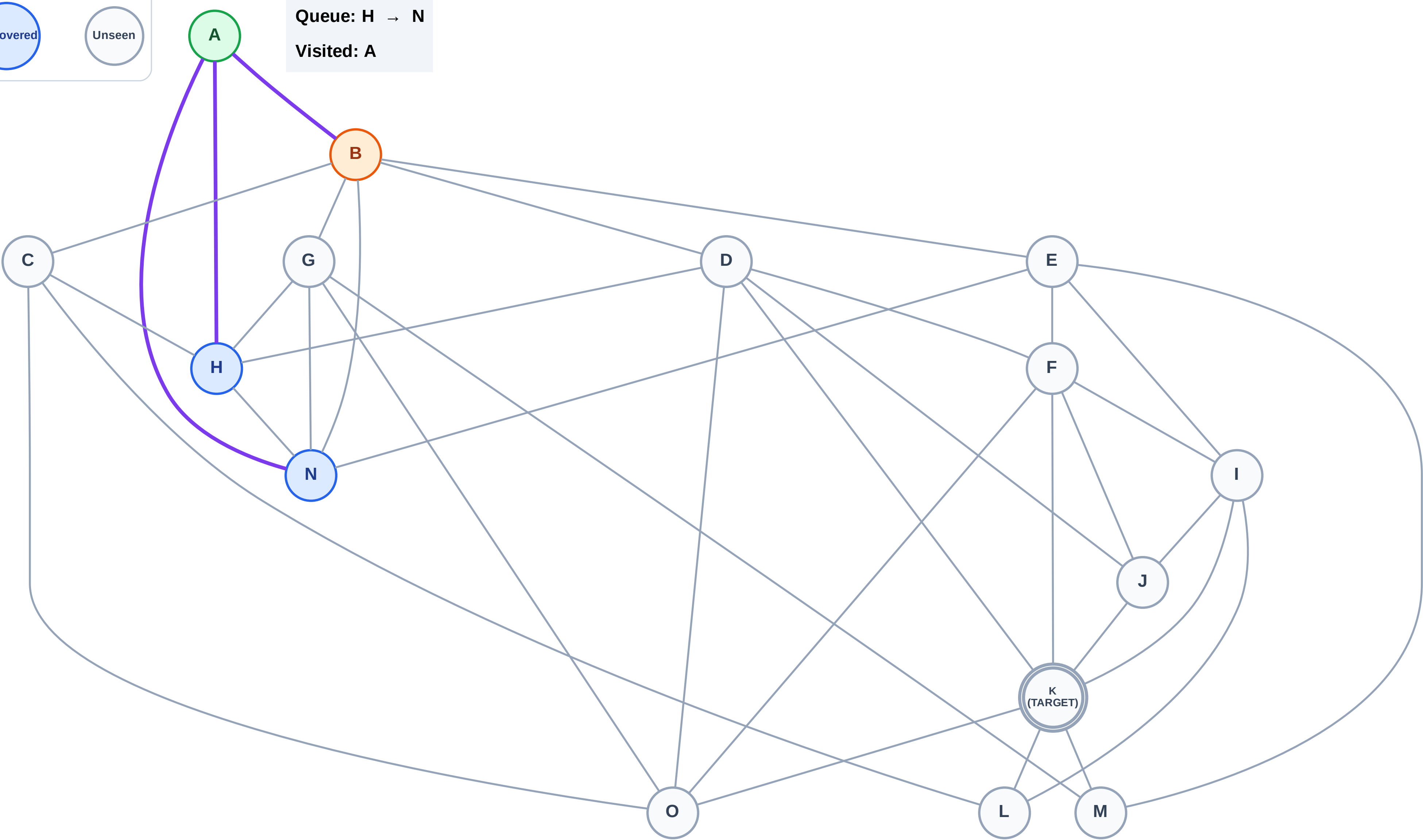
Processing

Discovered

Unseen

Queue: H → N

Visited: A



BFS Traversal

Step 5: Discover C, D, E, G from B

Legend

Complete

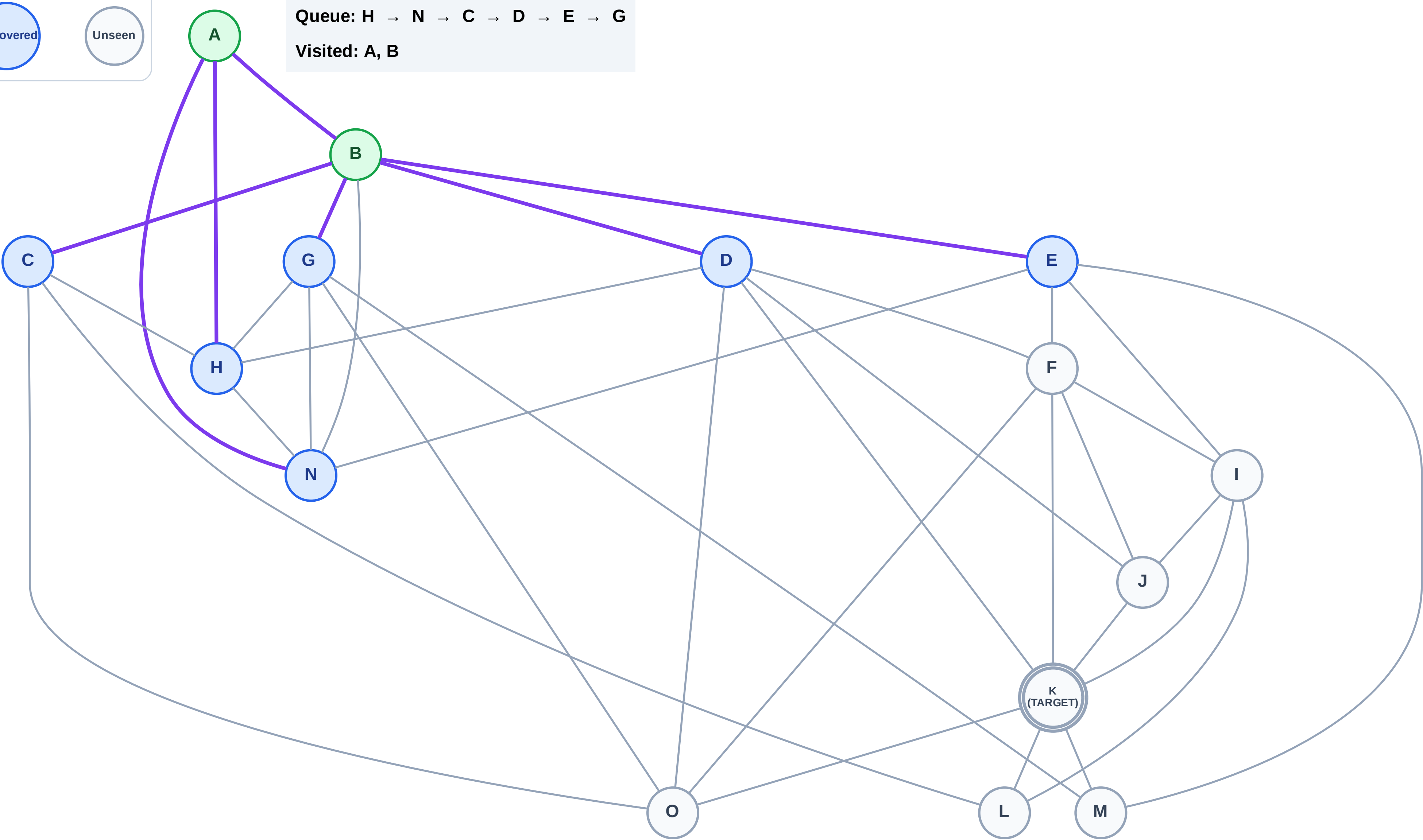
Processing

Discovered

Unseen

Queue: H → N → C → D → E → G

Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

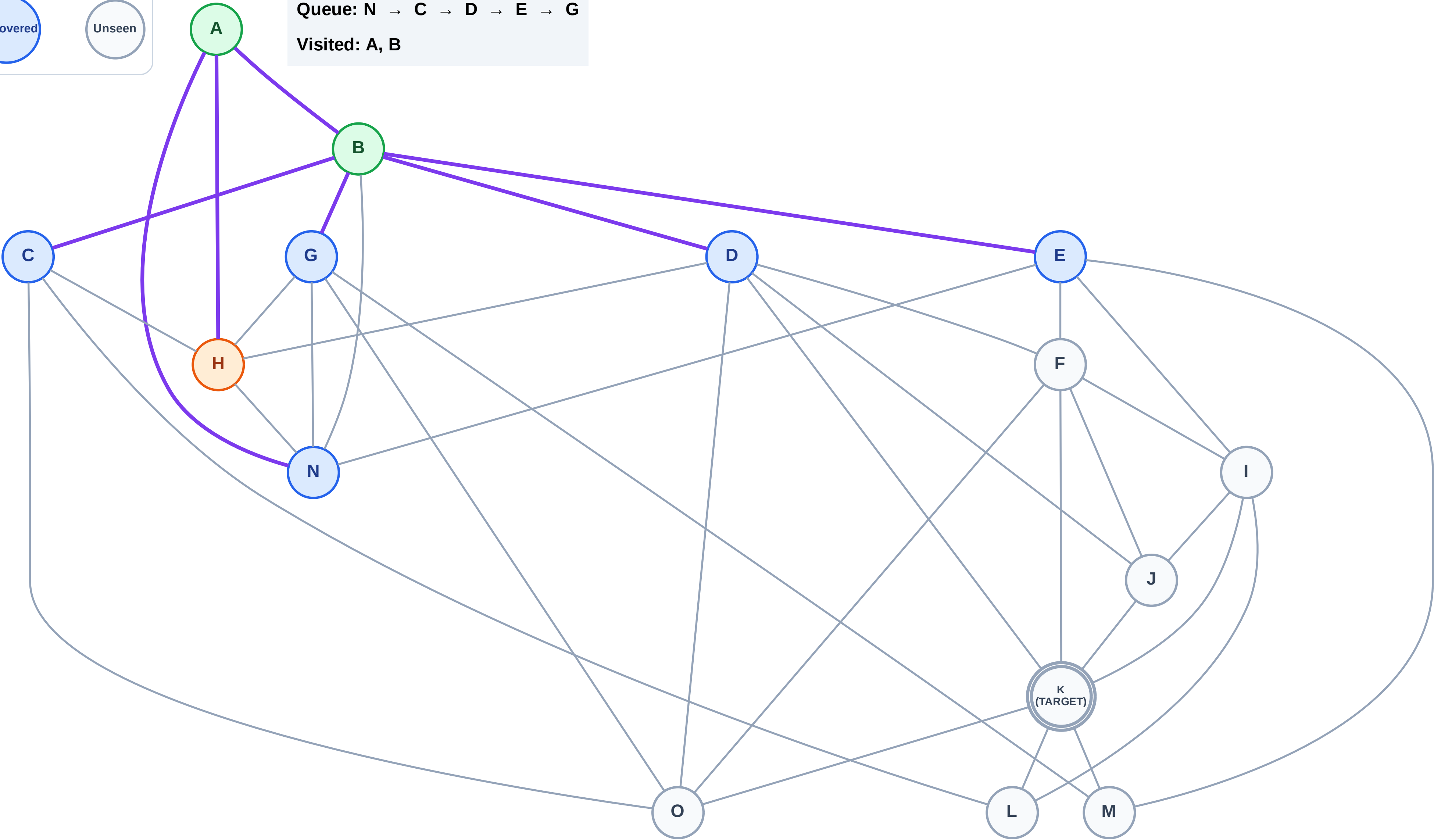
Processing

Discovered

Unseen

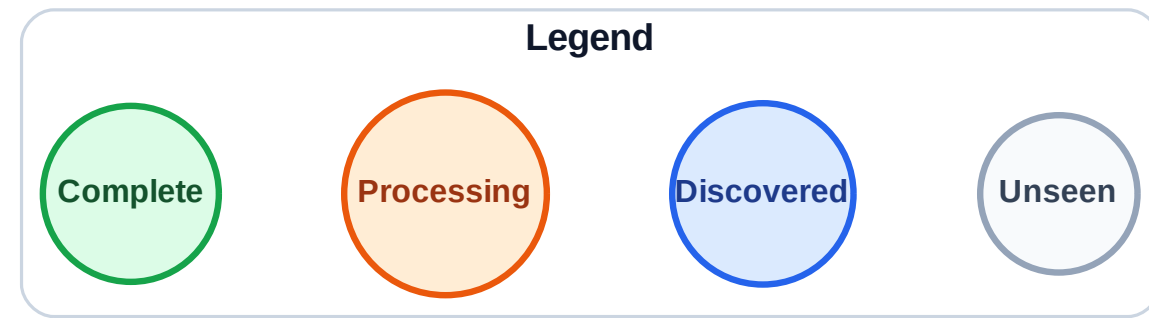
Queue: N → C → D → E → G

Visited: A, B

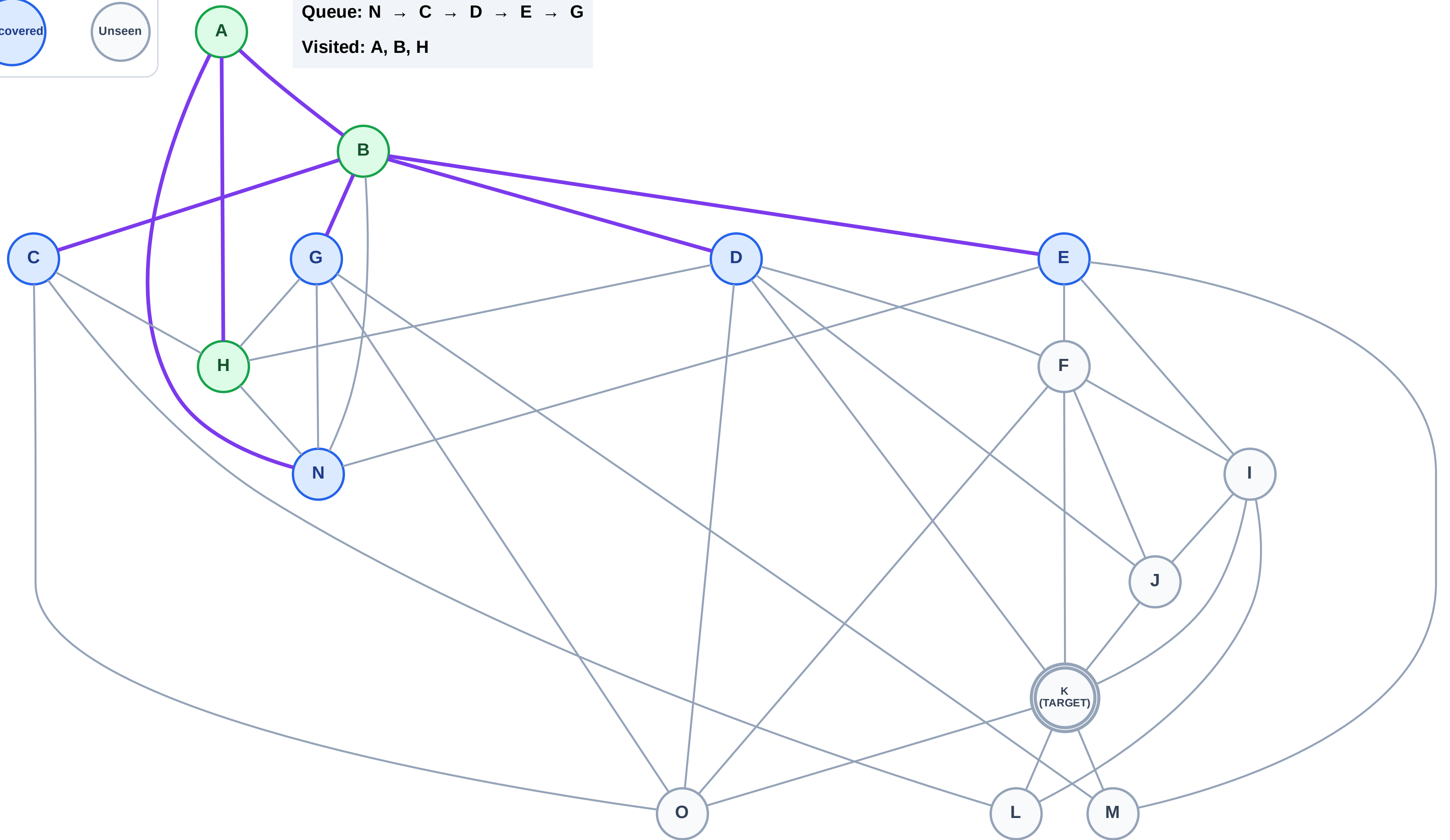


BFS Traversal

Step 7: No new neighbors from H



Queue: N → C → D → E → G
Visited: A, B, H



BFS Traversal

Step 8: Process node N

Legend

Complete

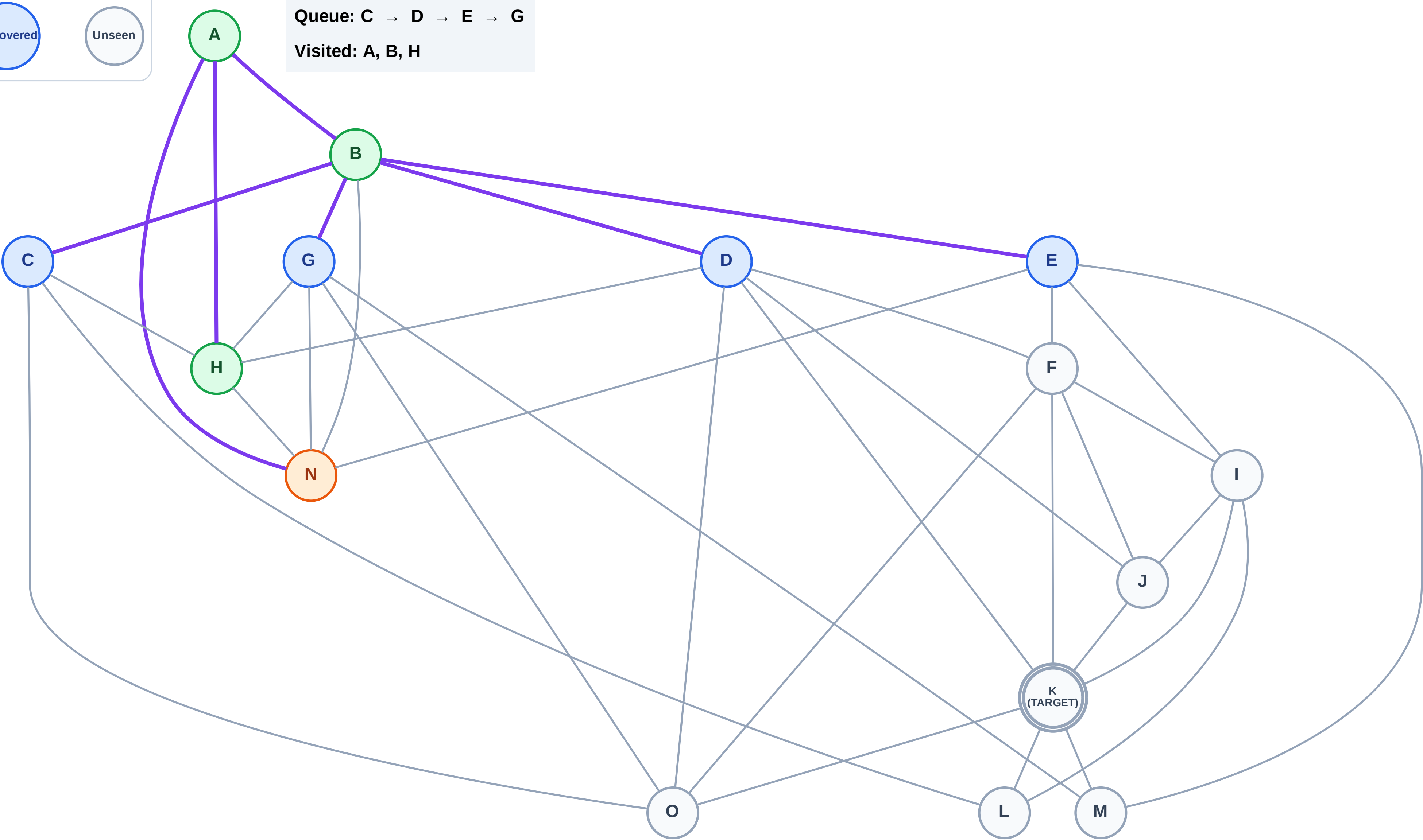
Processing

Discovered

Unseen

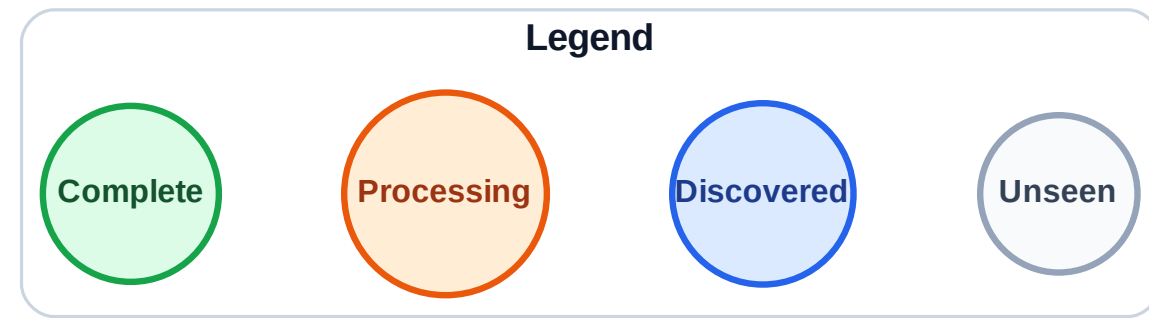
Queue: C → D → E → G

Visited: A, B, H



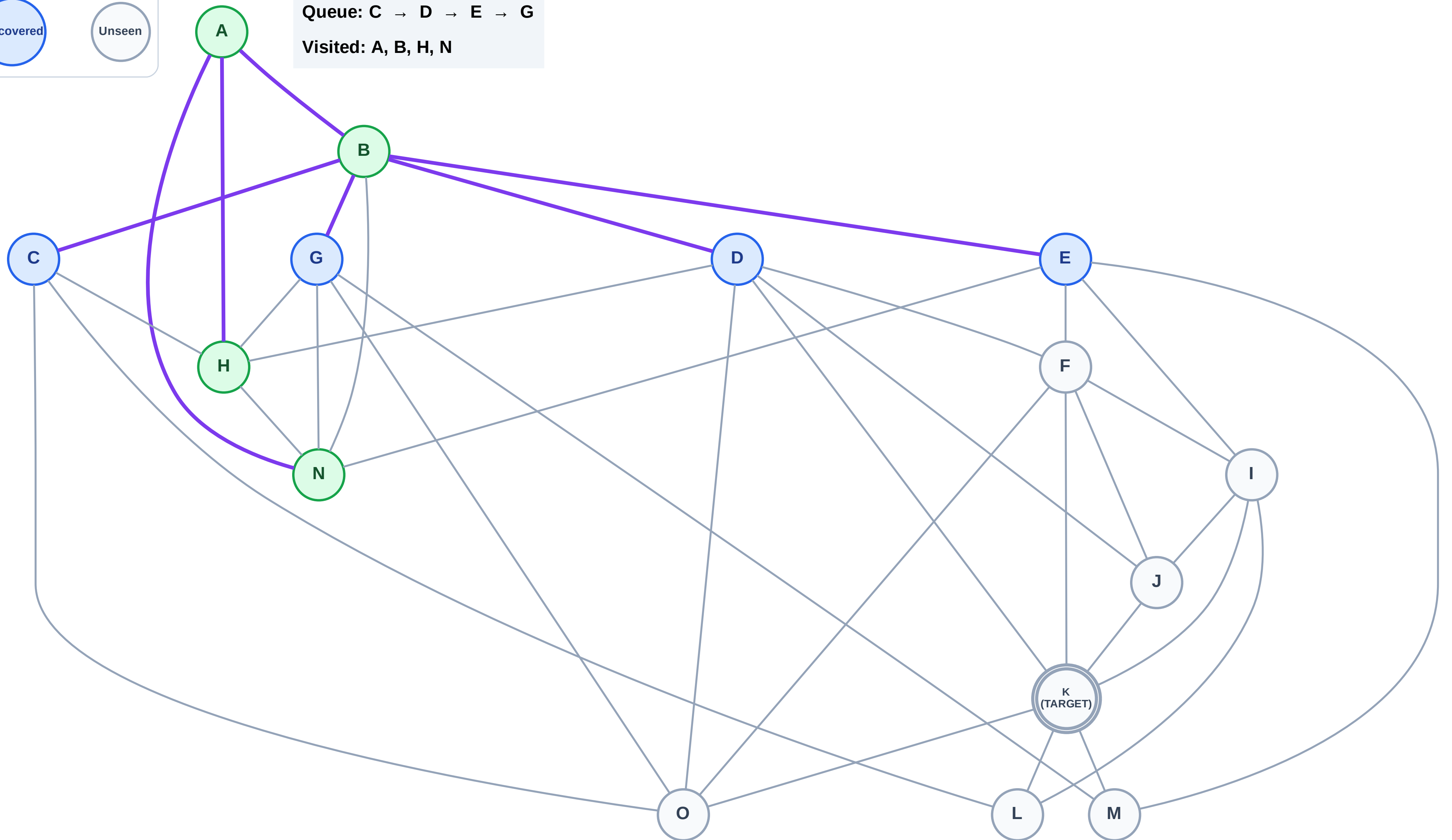
BFS Traversal

Step 9: No new neighbors from N



Queue: C → D → E → G

Visited: A, B, H, N



BFS Traversal

Step 10: Process node C

Legend

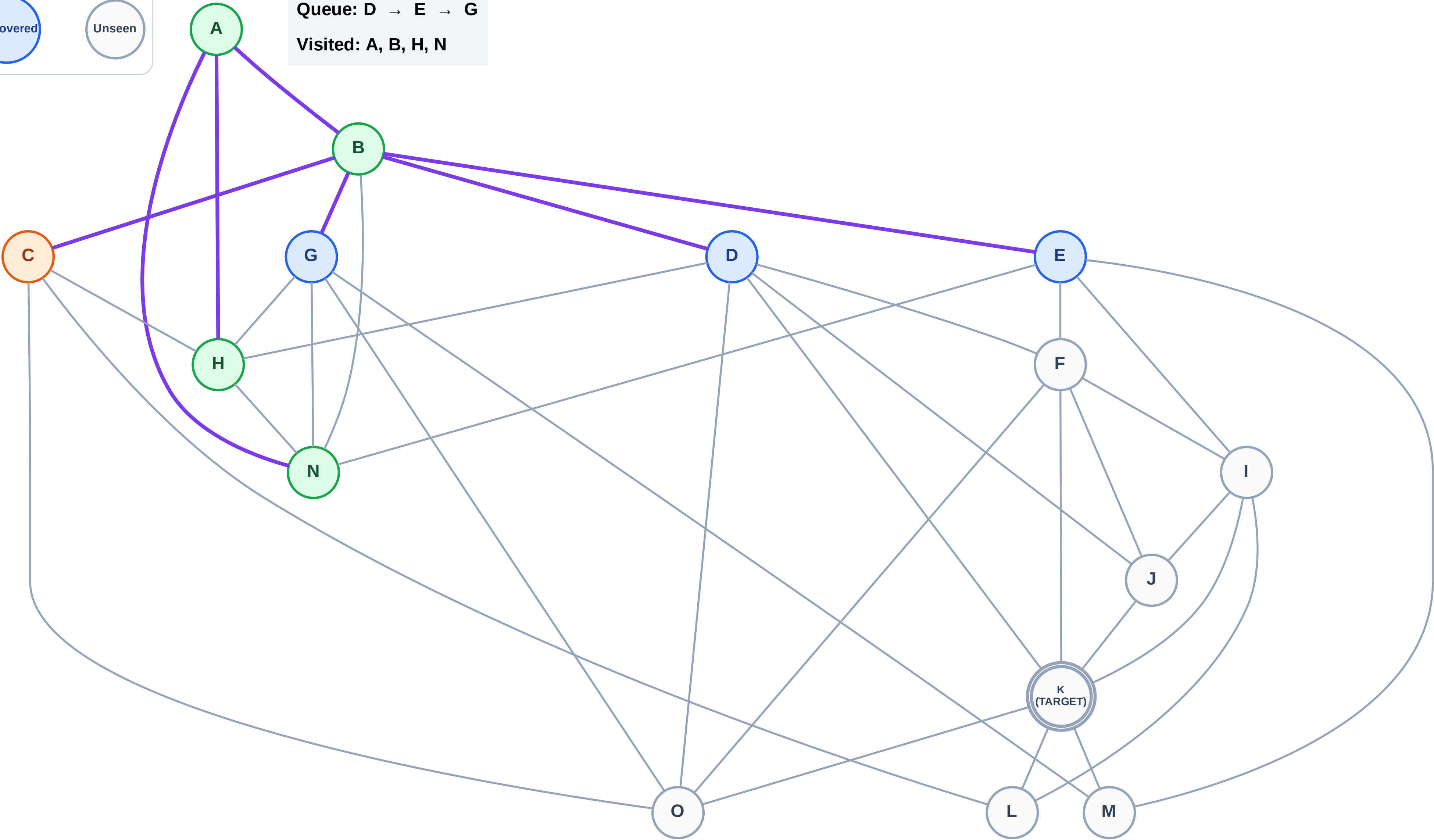
Complete

Processing

Discovered

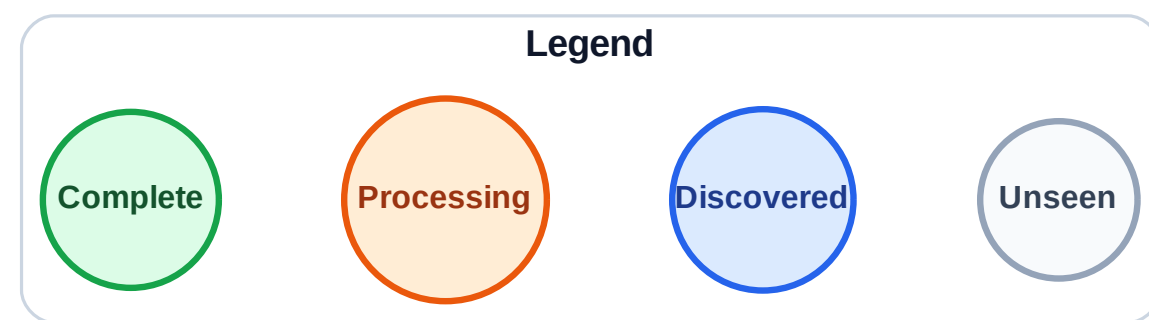
Unseen

Queue: D → E → G
Visited: A, B, H, N

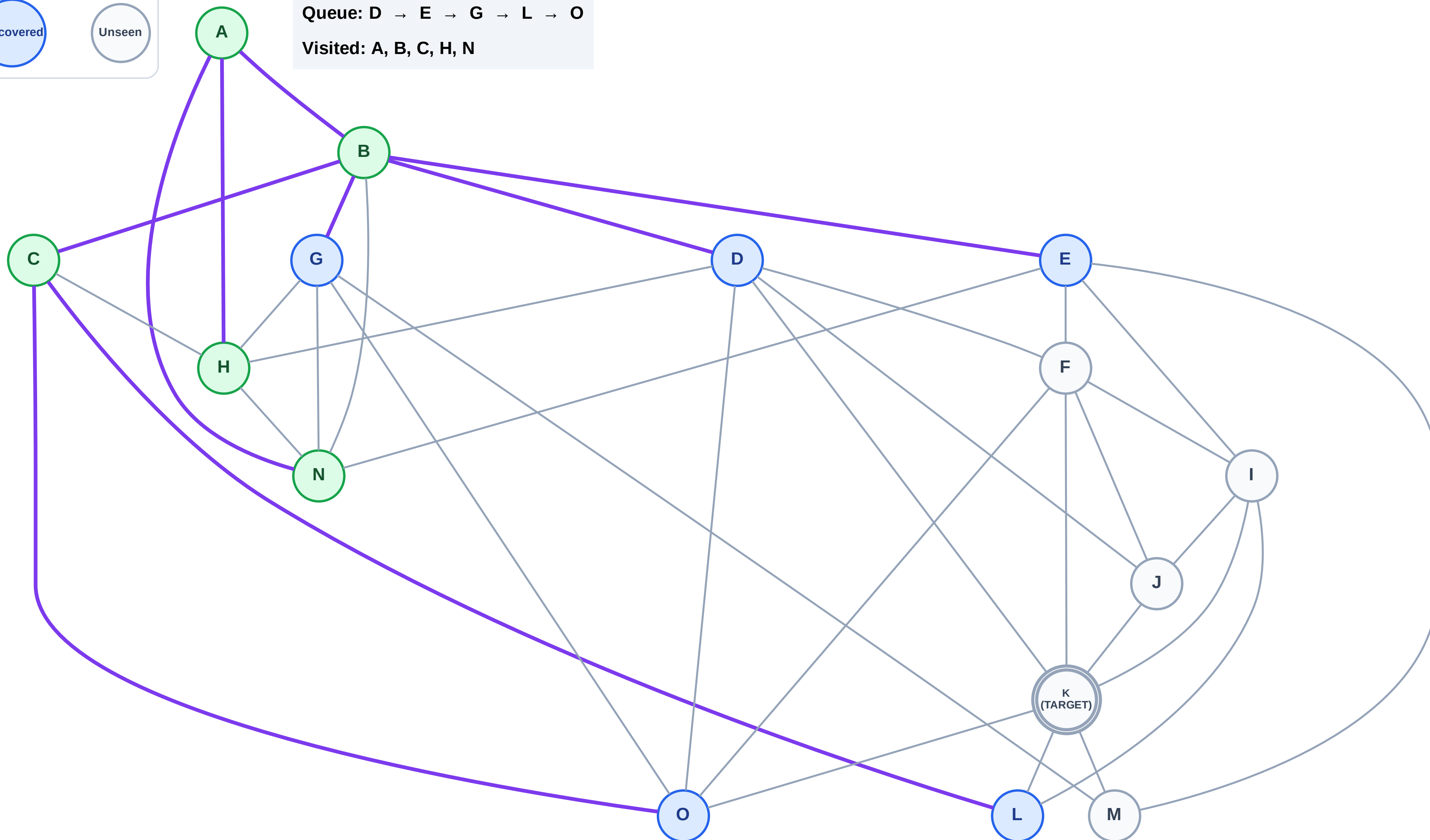


BFS Traversal

Step 11: Discover L, O from C



Queue: D → E → G → L → O
Visited: A, B, C, H, N



BFS Traversal

Step 12: Process node D

Legend

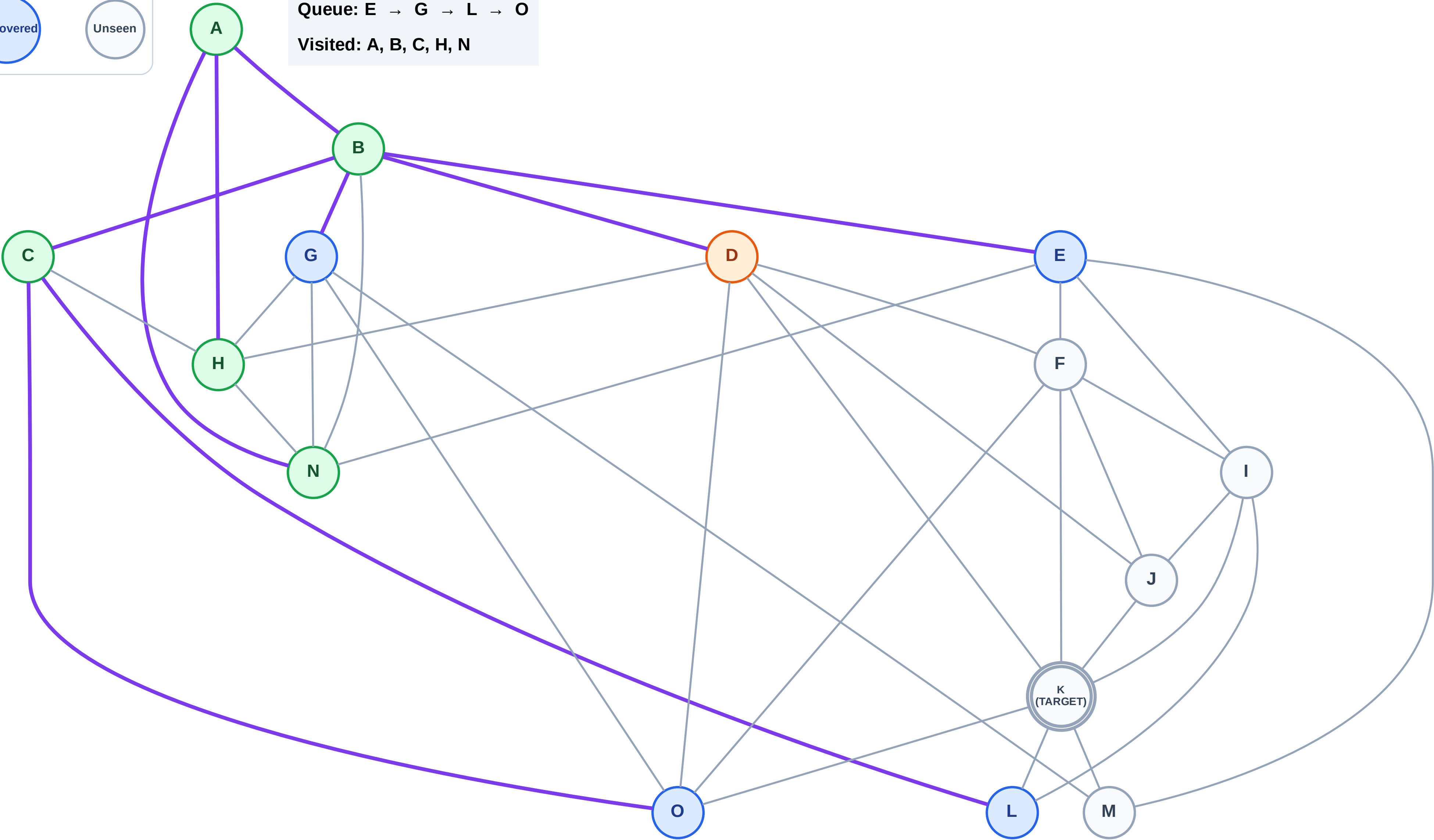
Complete

Processing

Discovered

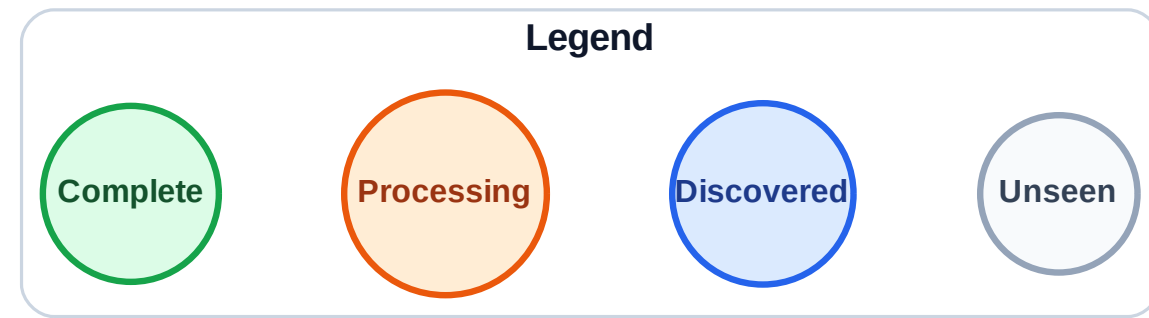
Unseen

Queue: E → G → L → O
Visited: A, B, C, H, N

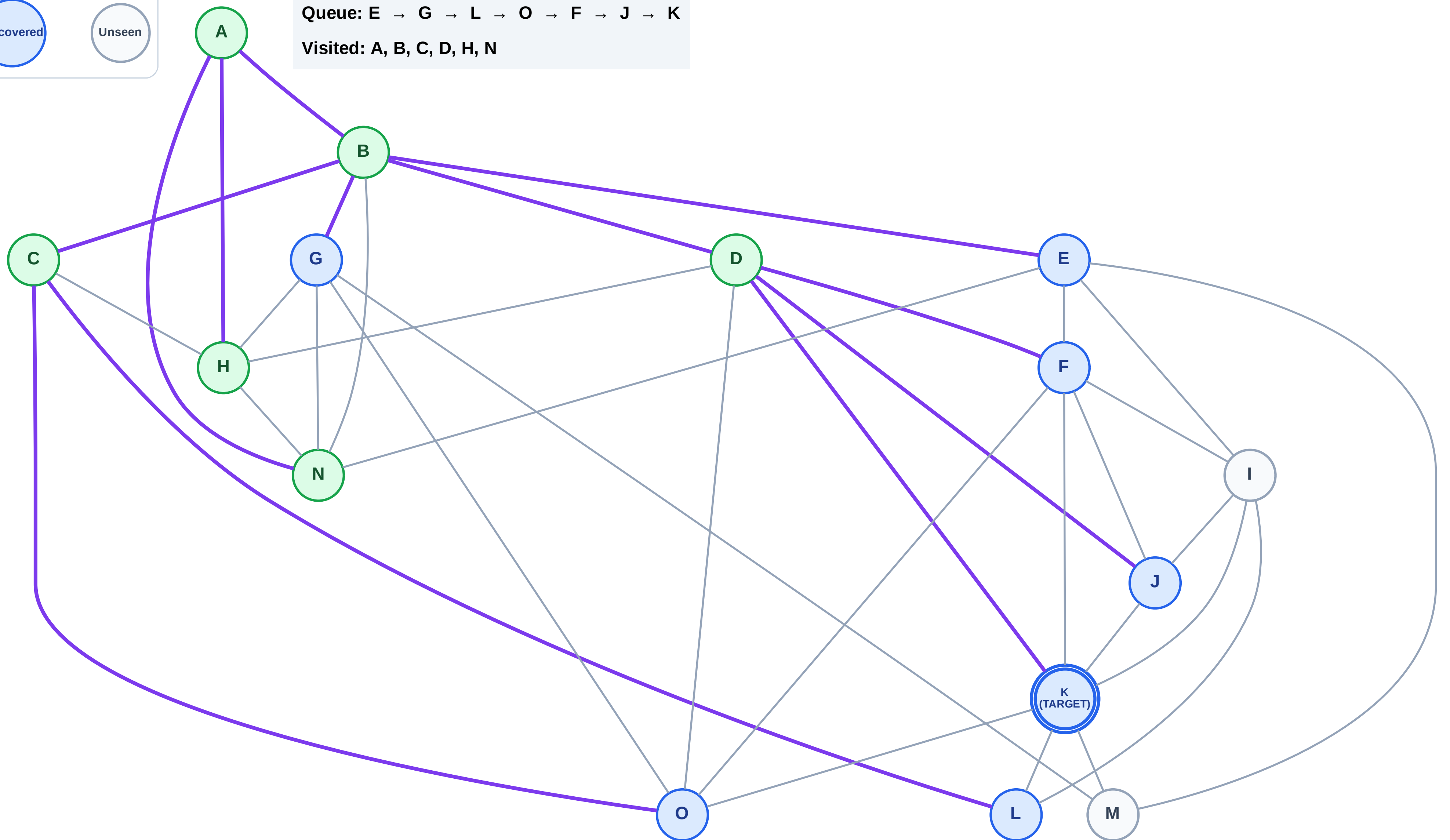


BFS Traversal

Step 13: Discover F, J, K from D



Queue: E → G → L → O → F → J → K
Visited: A, B, C, D, H, N



BFS Traversal

Step 14: Process node E

Legend

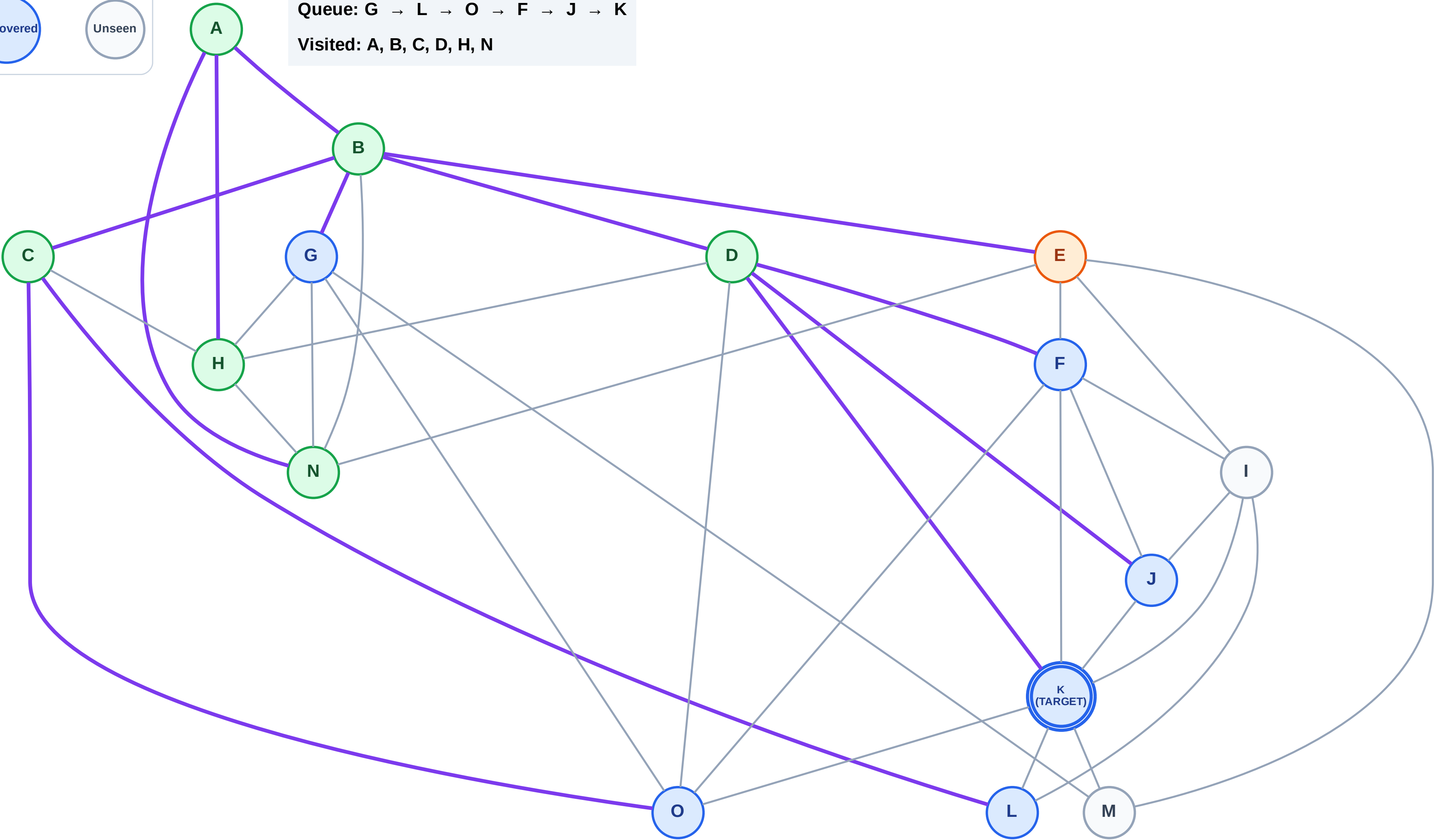
Complete

Processing

Discovered

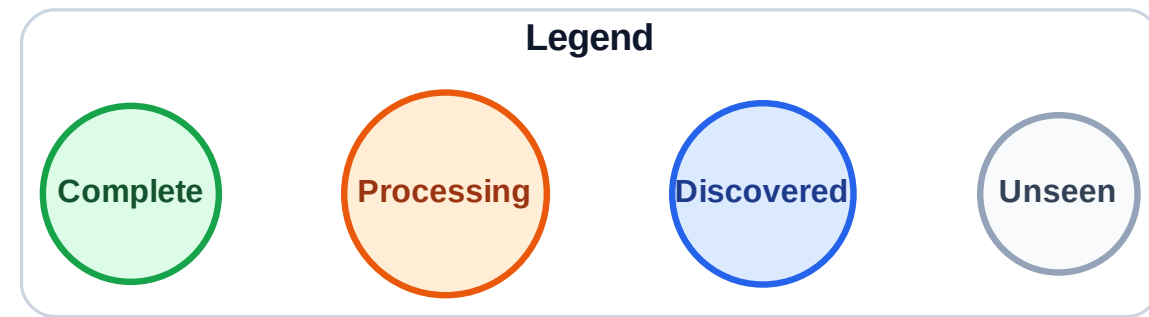
Unseen

Queue: G → L → O → F → J → K
Visited: A, B, C, D, H, N



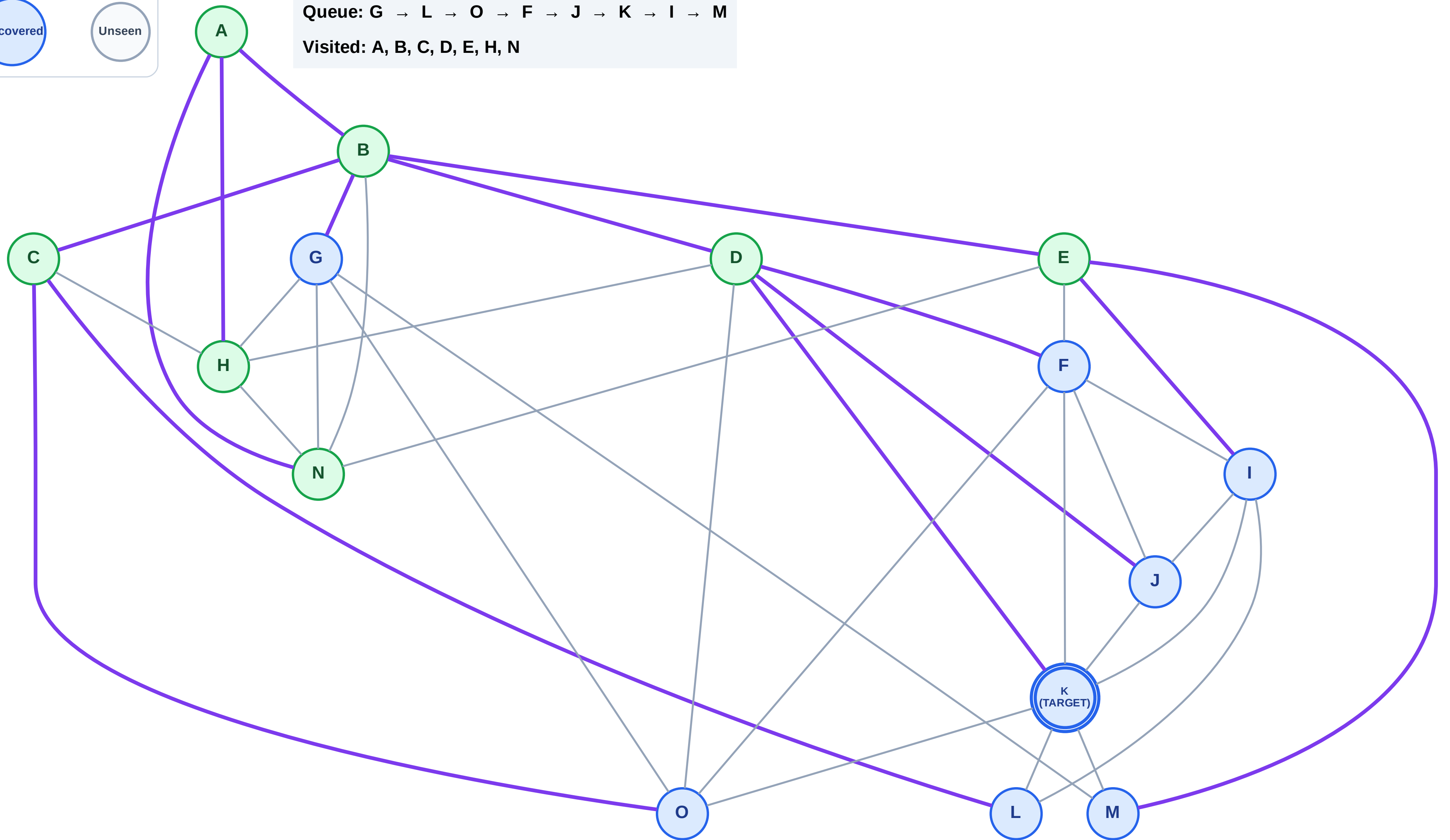
BFS Traversal

Step 15: Discover I, M from E



Queue: G → L → O → F → J → K → I → M

Visited: A, B, C, D, E, H, N



BFS Traversal

Step 16: Process node G

Legend

Complete

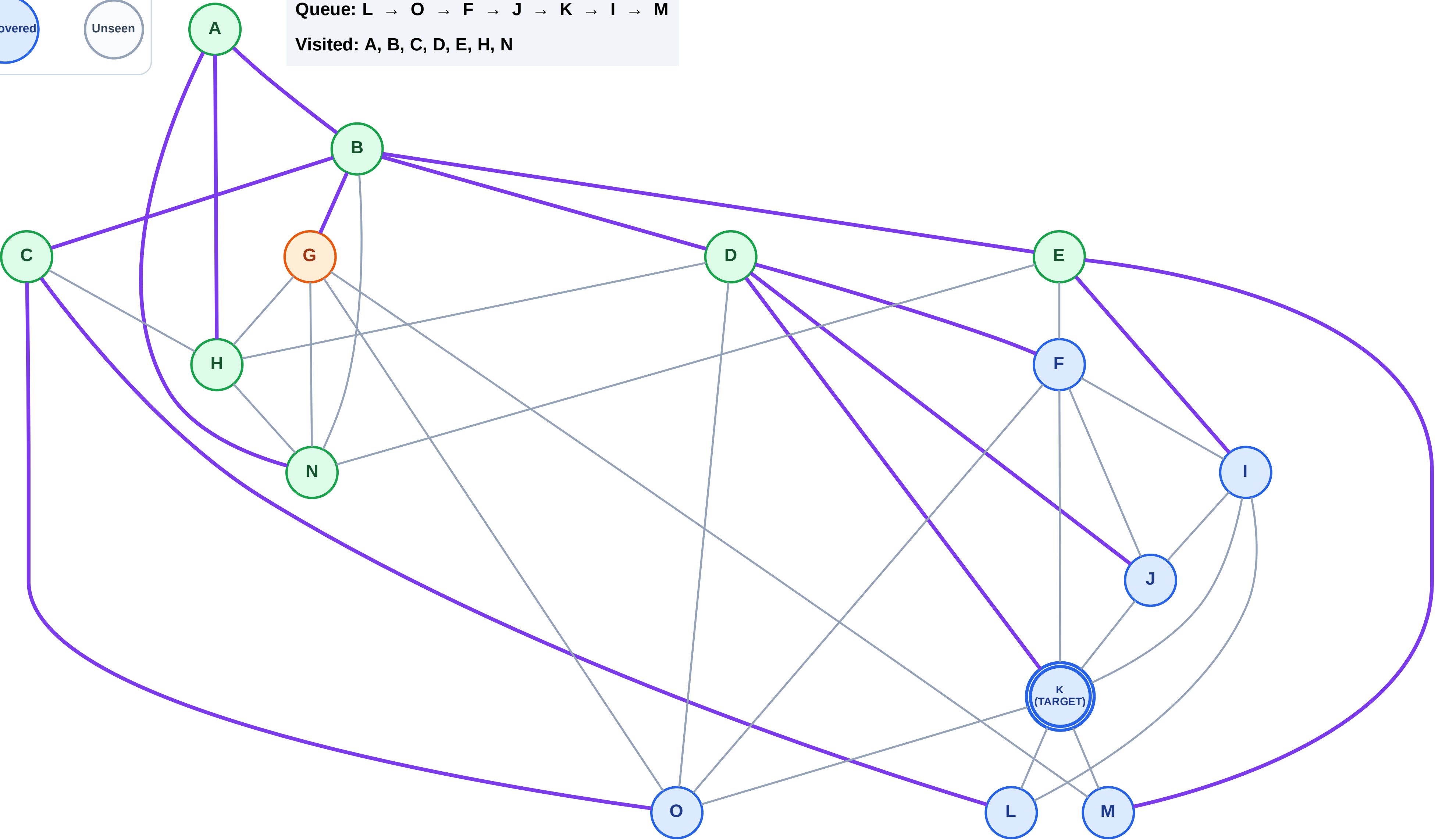
Processing

Discovered

Unseen

Queue: L → O → F → J → K → I → M

Visited: A, B, C, D, E, H, N



BFS Traversal

Step 17: No new neighbors from G

Legend

Complete

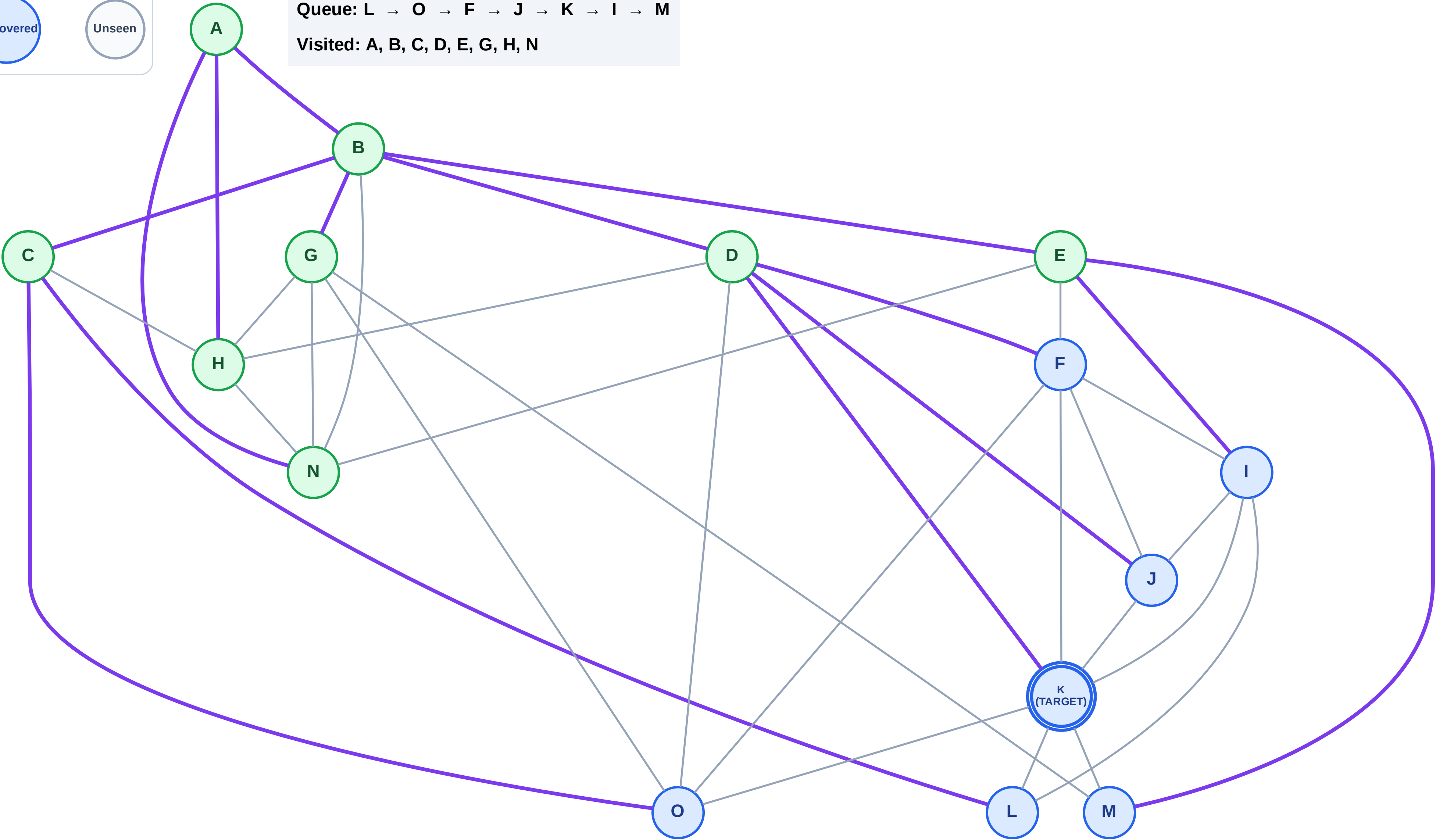
Processing

Discovered

Unseen

Queue: L → O → F → J → K → I → M

Visited: A, B, C, D, E, G, H, N



BFS Traversal

Step 18: Process node L

Legend

Complete

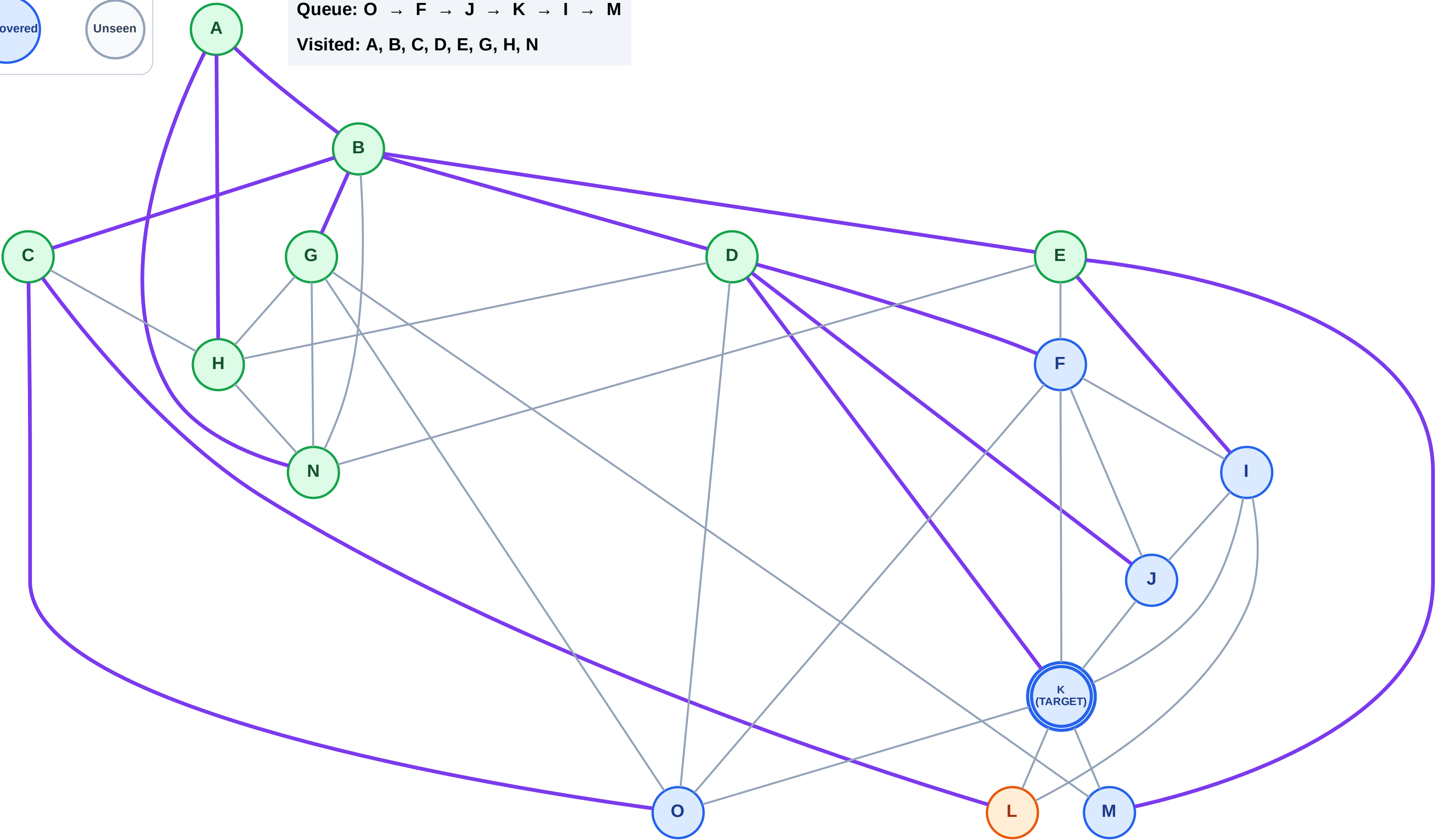
Processing

Discovered

Unseen

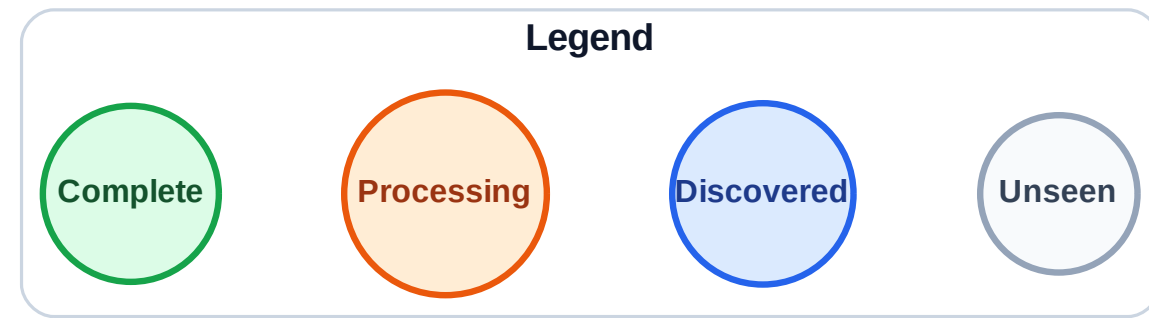
Queue: O → F → J → K → I → M

Visited: A, B, C, D, E, G, H, N

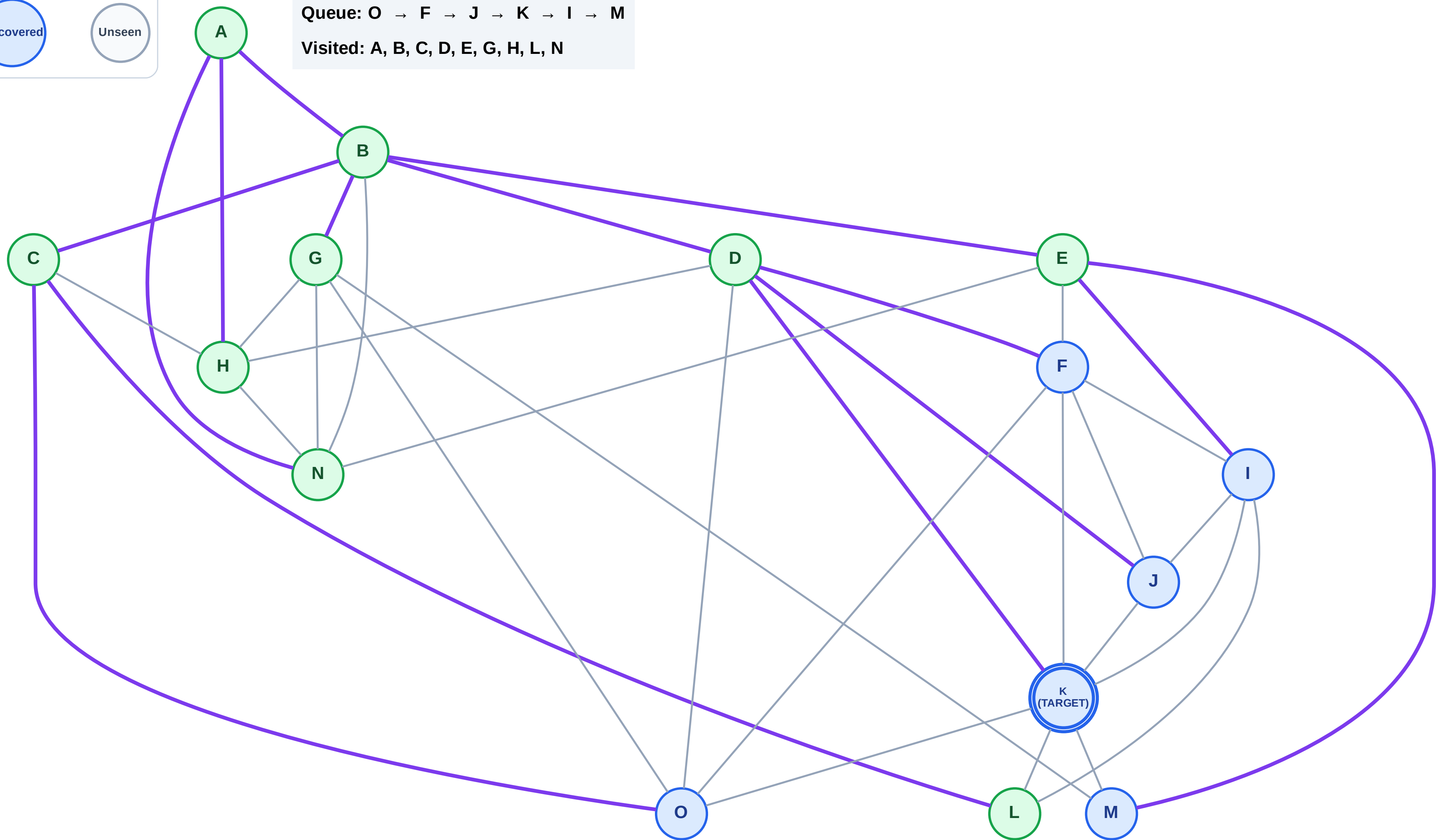


BFS Traversal

Step 19: No new neighbors from L



Queue: O → F → J → K → I → M
Visited: A, B, C, D, E, G, H, L, N



BFS Traversal

Step 20: Process node O

Legend

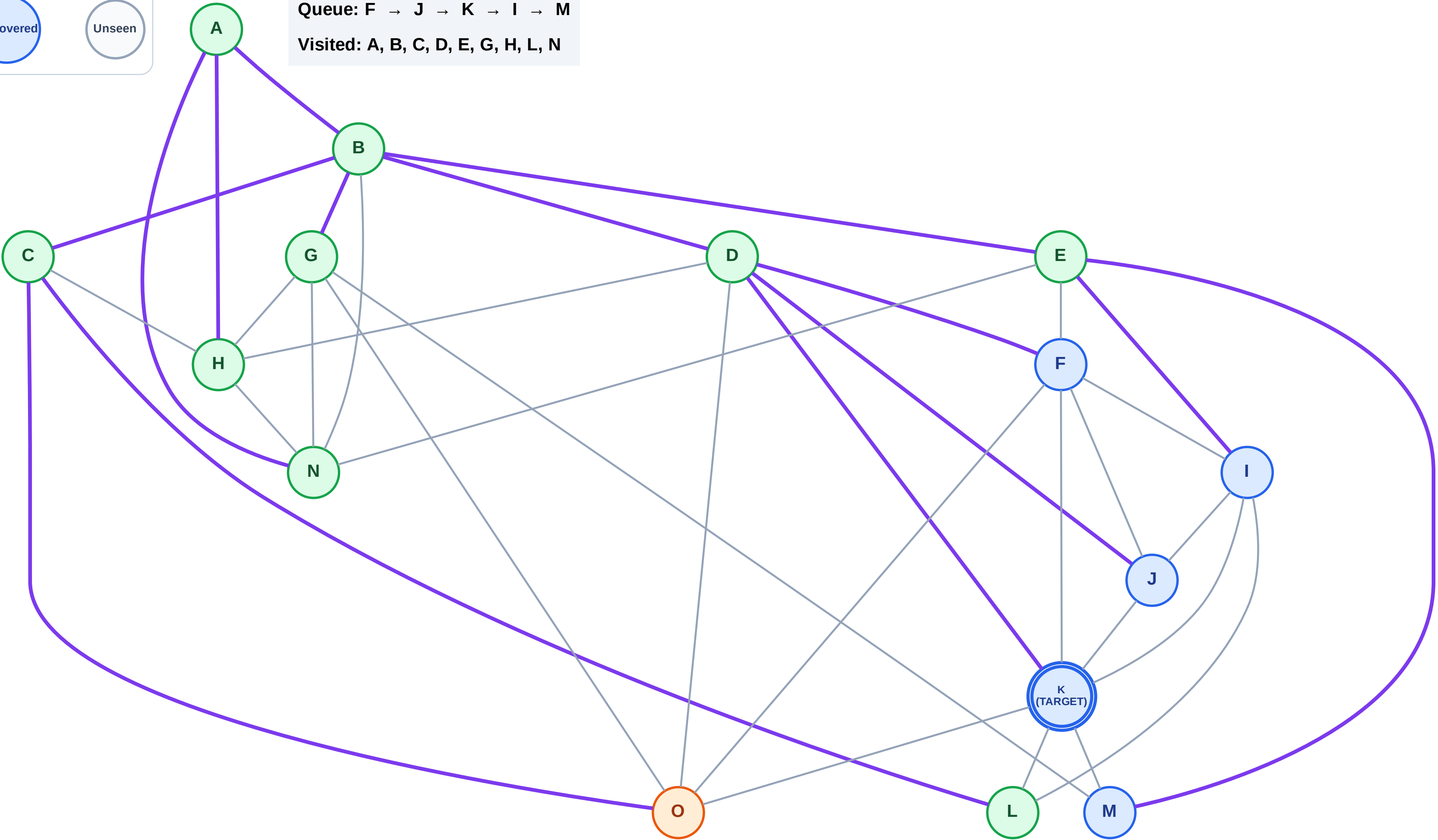
Complete

Processing

Discovered

Unseen

Queue: F → J → K → I → M
Visited: A, B, C, D, E, G, H, L, N



BFS Traversal

Step 21: No new neighbors from O

Legend

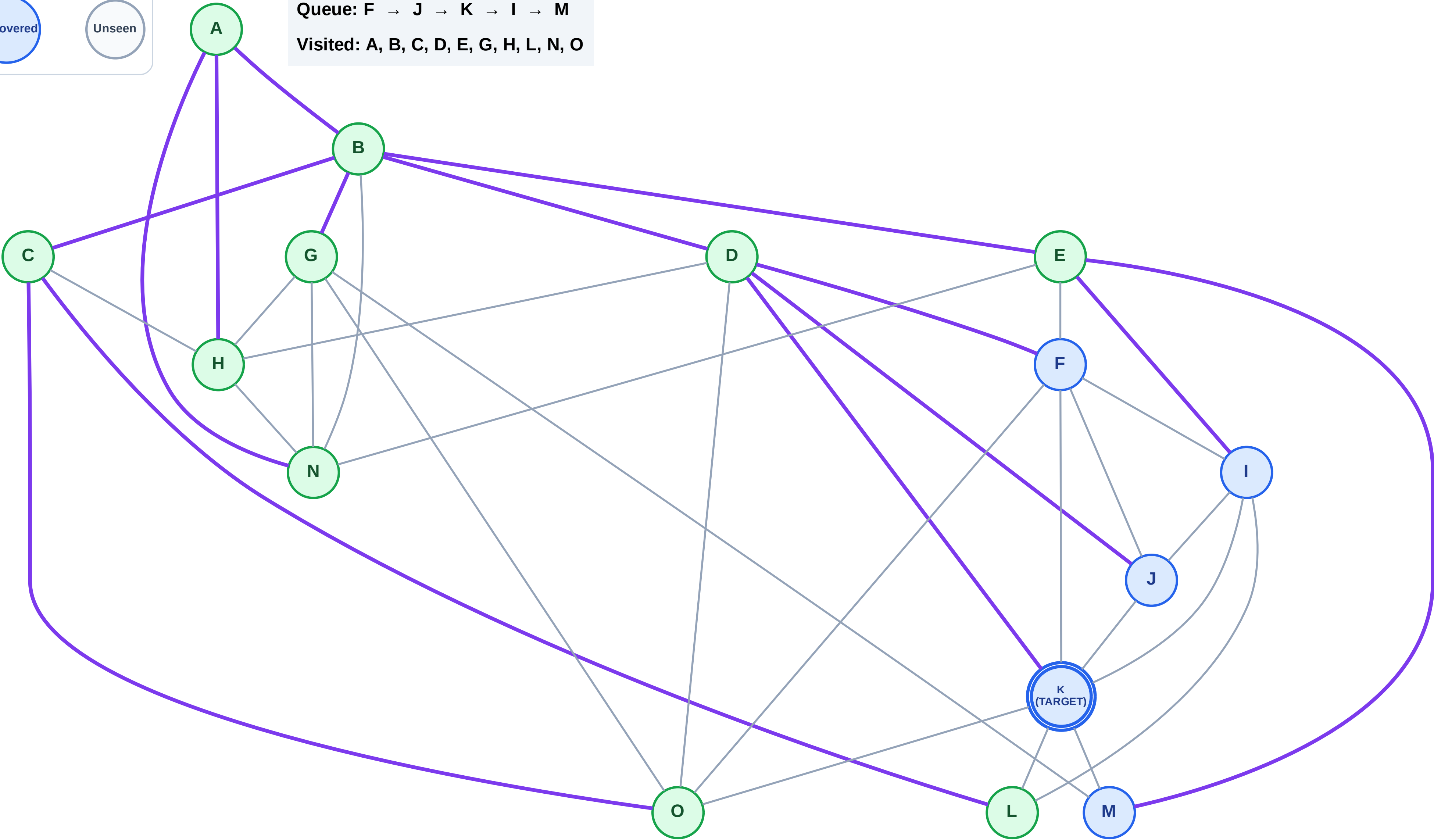
Complete

Processing

Discovered

Unseen

Queue: F → J → K → I → M
Visited: A, B, C, D, E, G, H, L, N, O



BFS Traversal

Step 22: Process node F

Legend

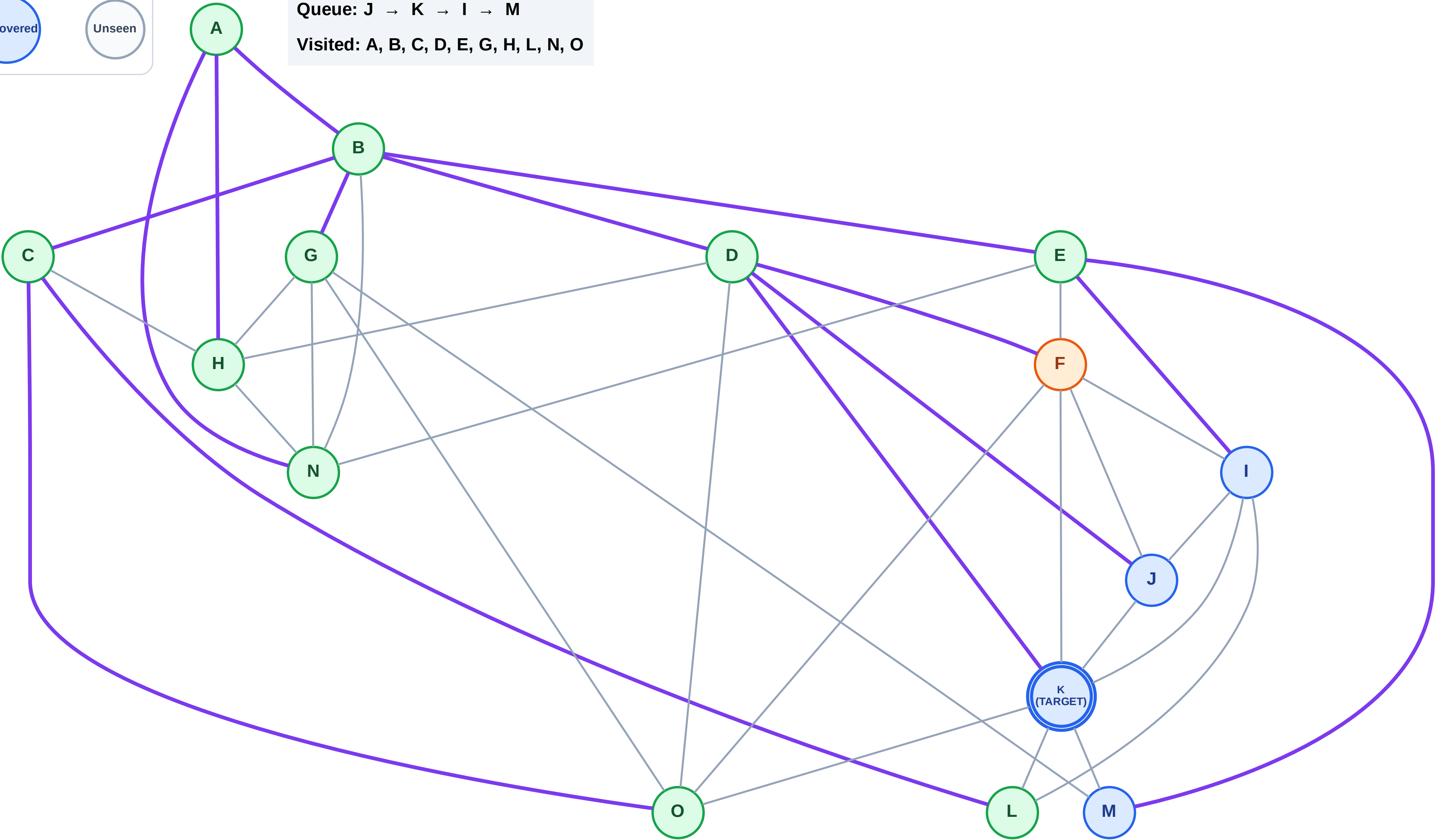
Complete

Processing

Discovered

Unseen

Queue: J → K → I → M
Visited: A, B, C, D, E, G, H, L, N, O



BFS Traversal

Step 23: No new neighbors from F

Legend

Complete

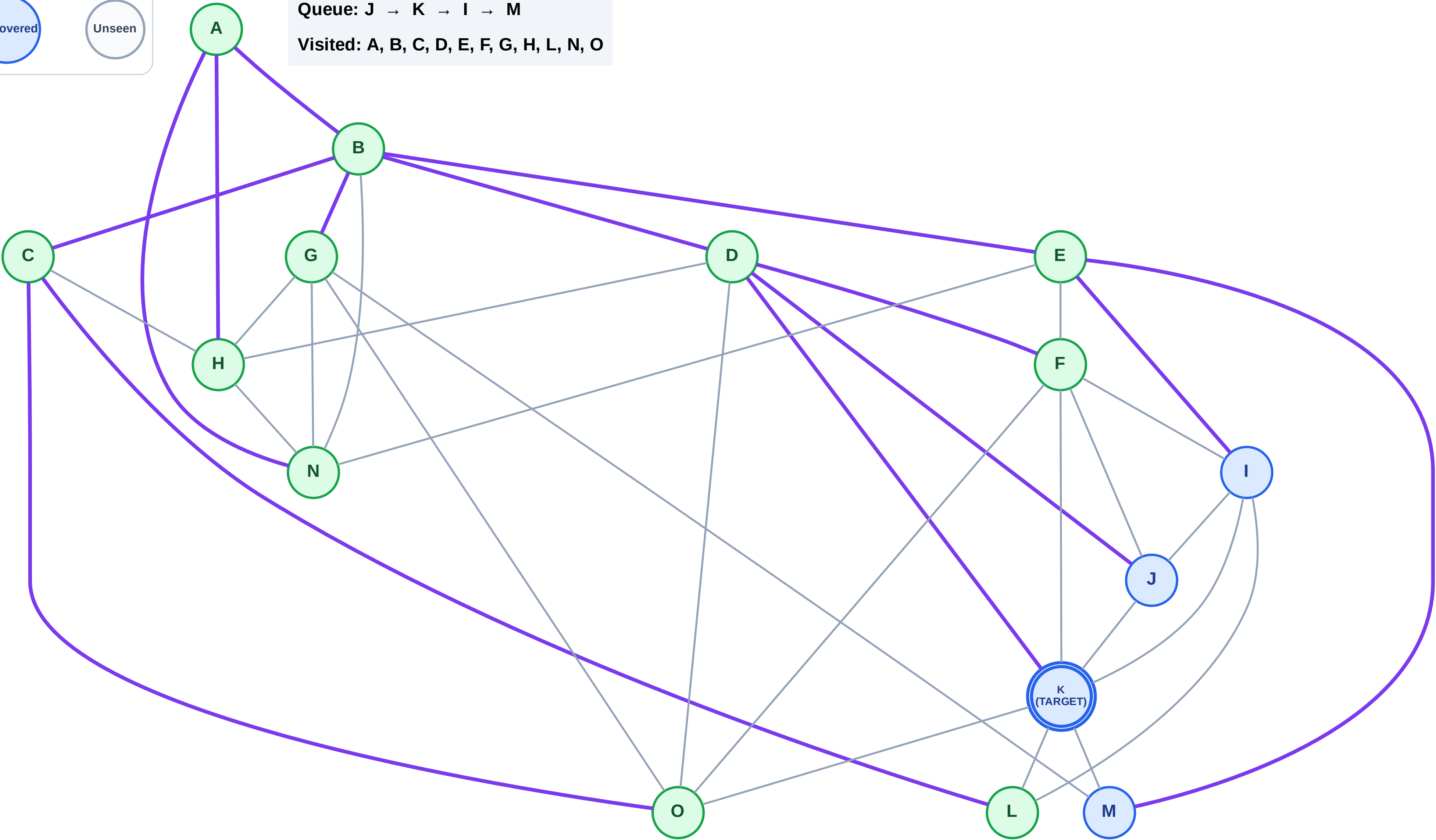
Processing

Discovered

Unseen

Queue: J → K → I → M

Visited: A, B, C, D, E, F, G, H, L, N, O



BFS Traversal

Step 24: Process node J

Legend

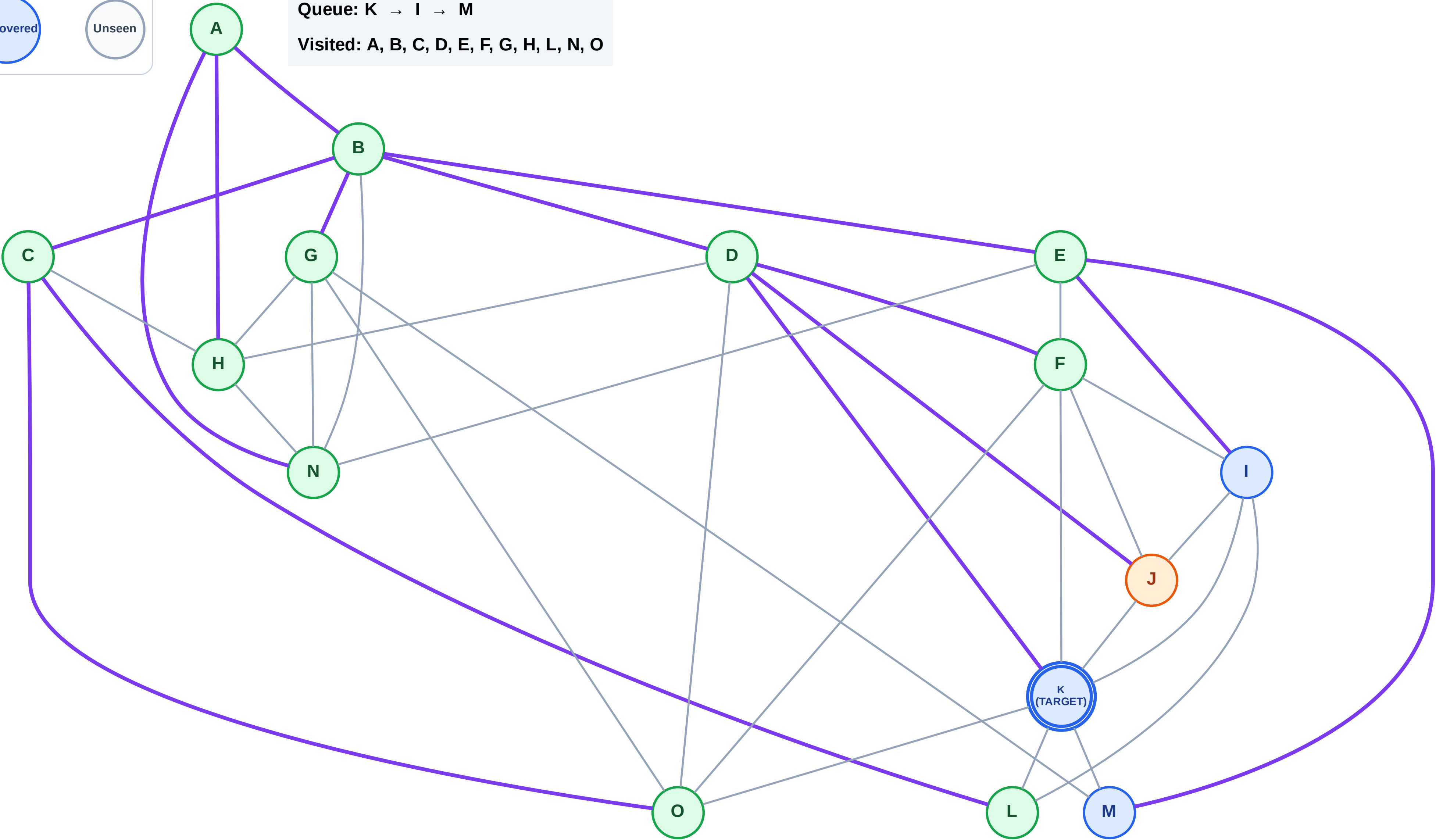
Complete

Processing

Discovered

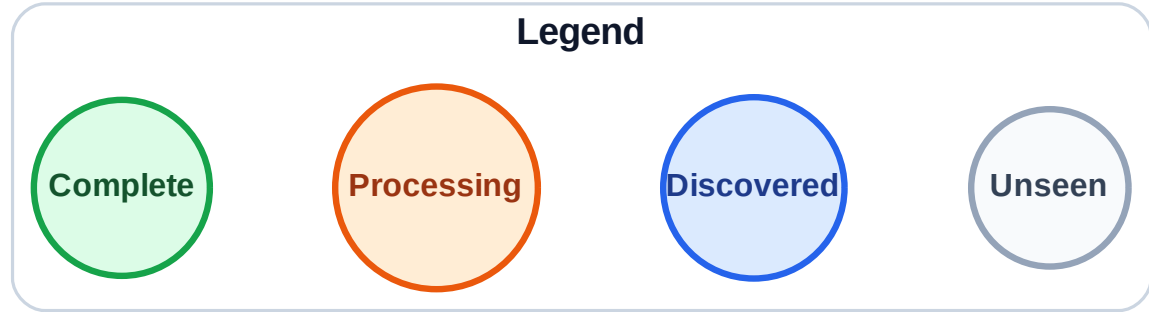
Unseen

Queue: K → I → M
Visited: A, B, C, D, E, F, G, H, L, N, O



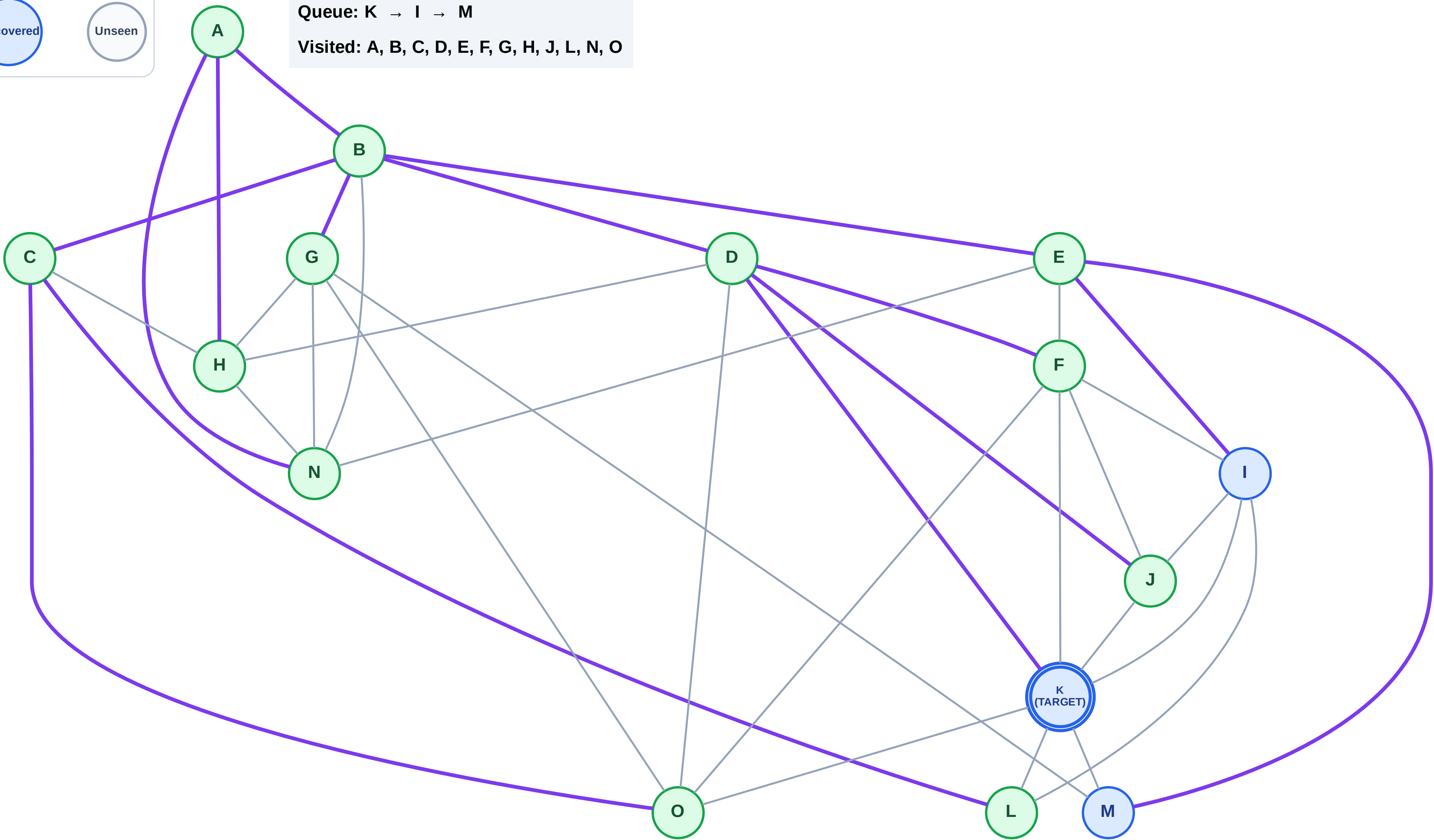
BFS Traversal

Step 25: No new neighbors from J



Queue: K → I → M

Visited: A, B, C, D, E, F, G, H, J, L, N, O



BFS Traversal

Step 26: Process node K

Legend

Complete

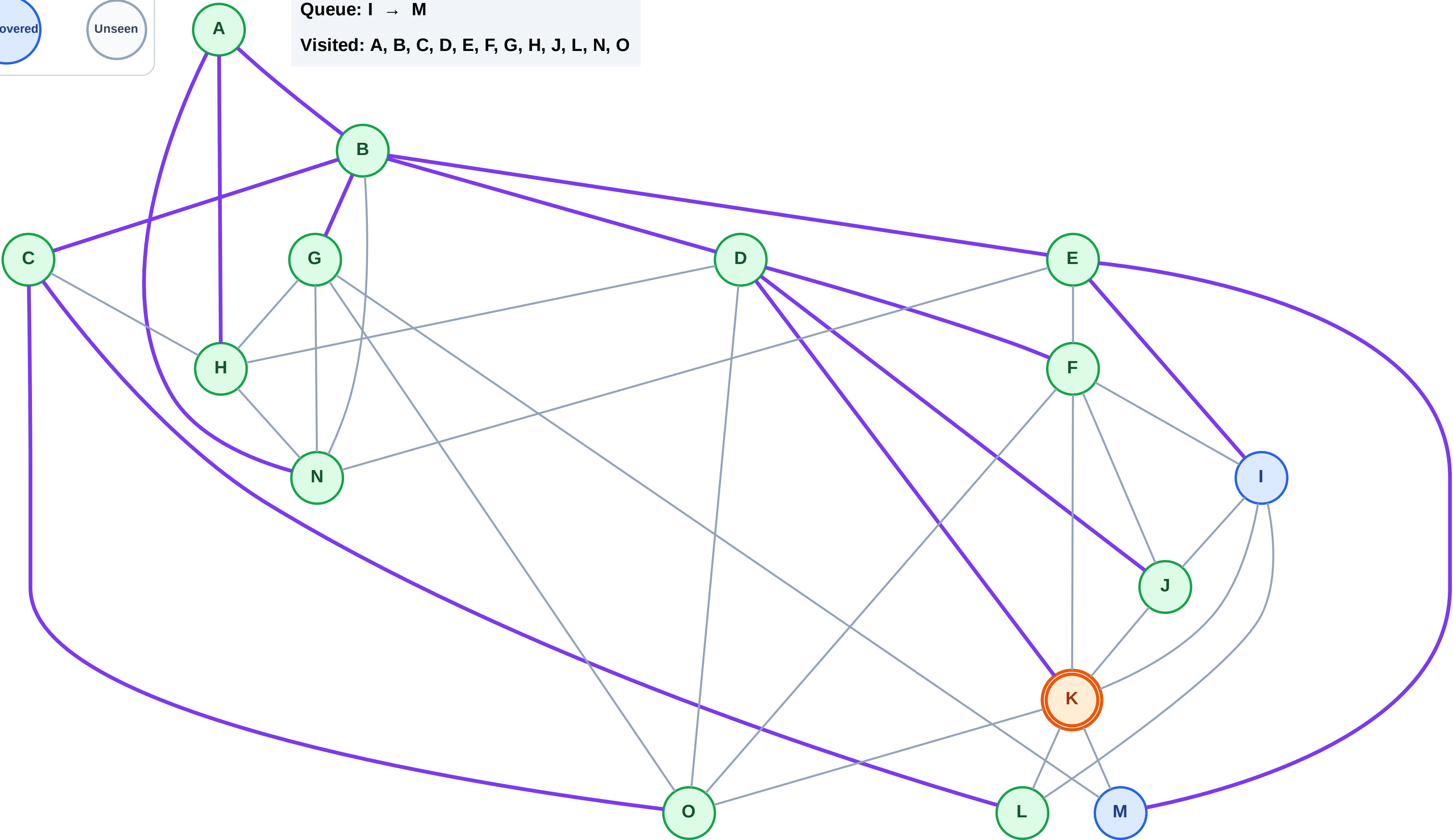
Processing

Discovered

Unseen

Queue: I → M

Visited: A, B, C, D, E, F, G, H, J, L, N, O



BFS Traversal

Step 27: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

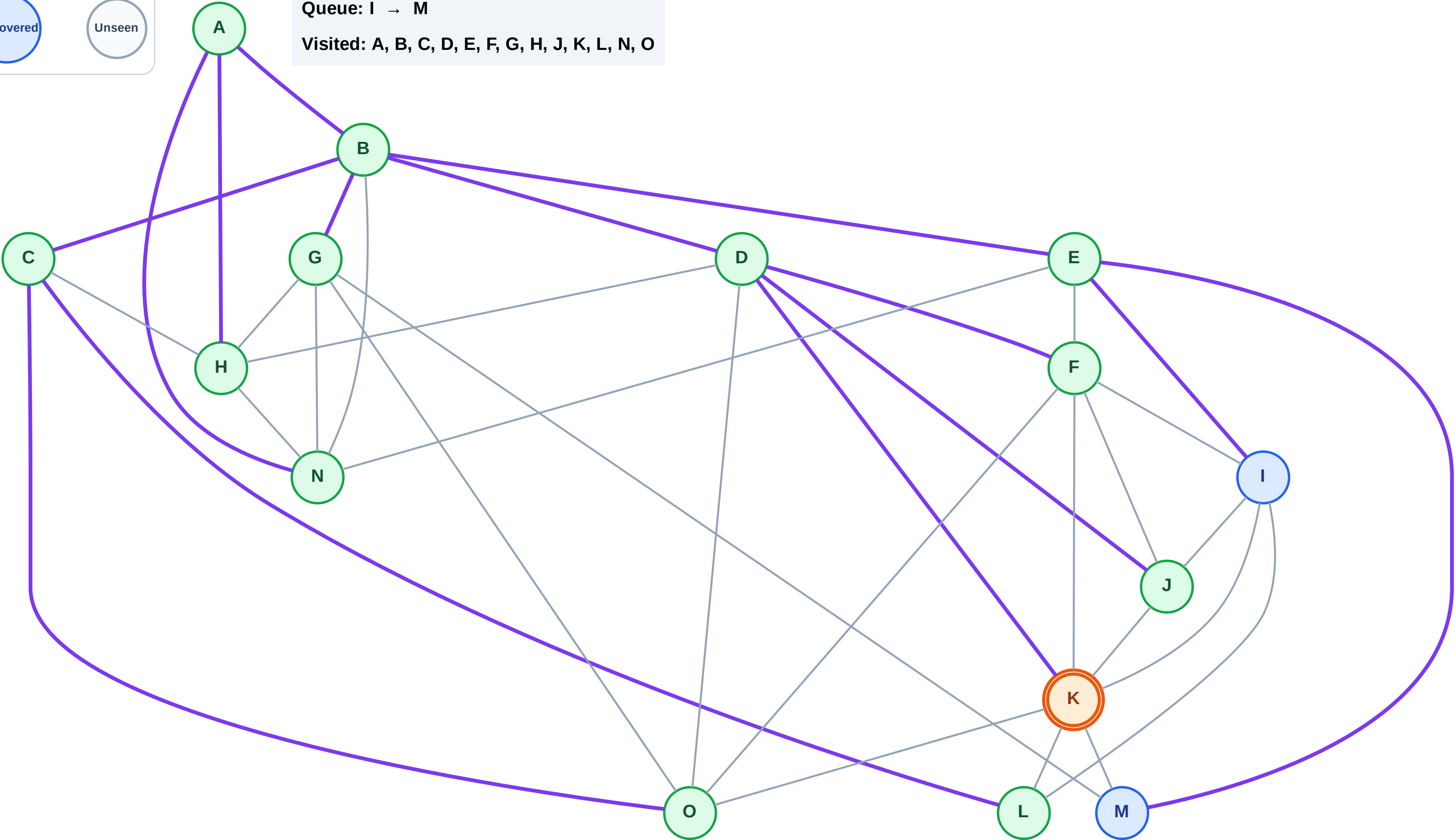
Processing

Discovered

Unseen

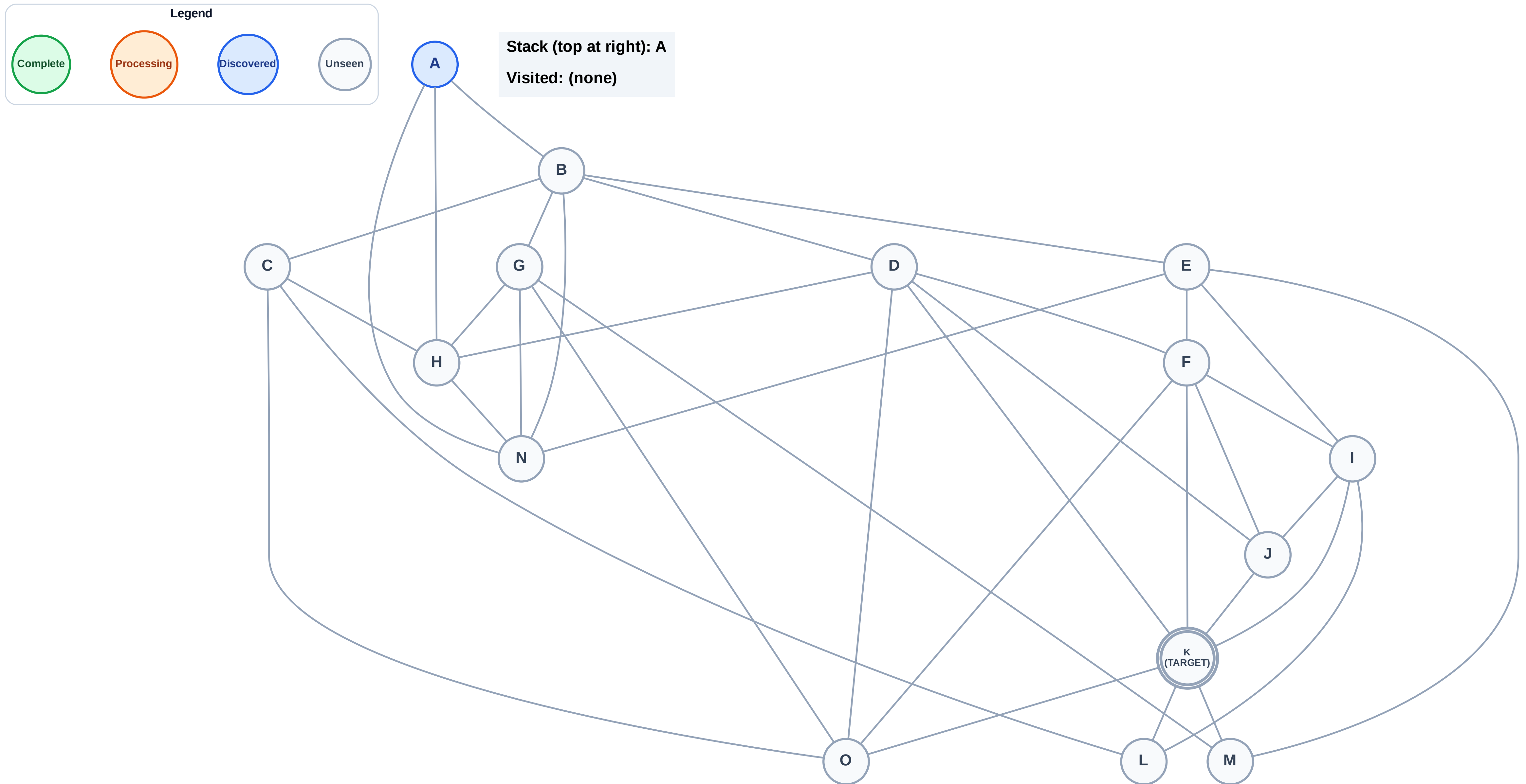
Queue: I → M

Visited: A, B, C, D, E, F, G, H, J, K, L, N, O



Step 1: Start at A (Seeking target node K)

Step 1: Start at A (Seeking target node K)



DFS Traversal

Step 2: Process node A

Legend

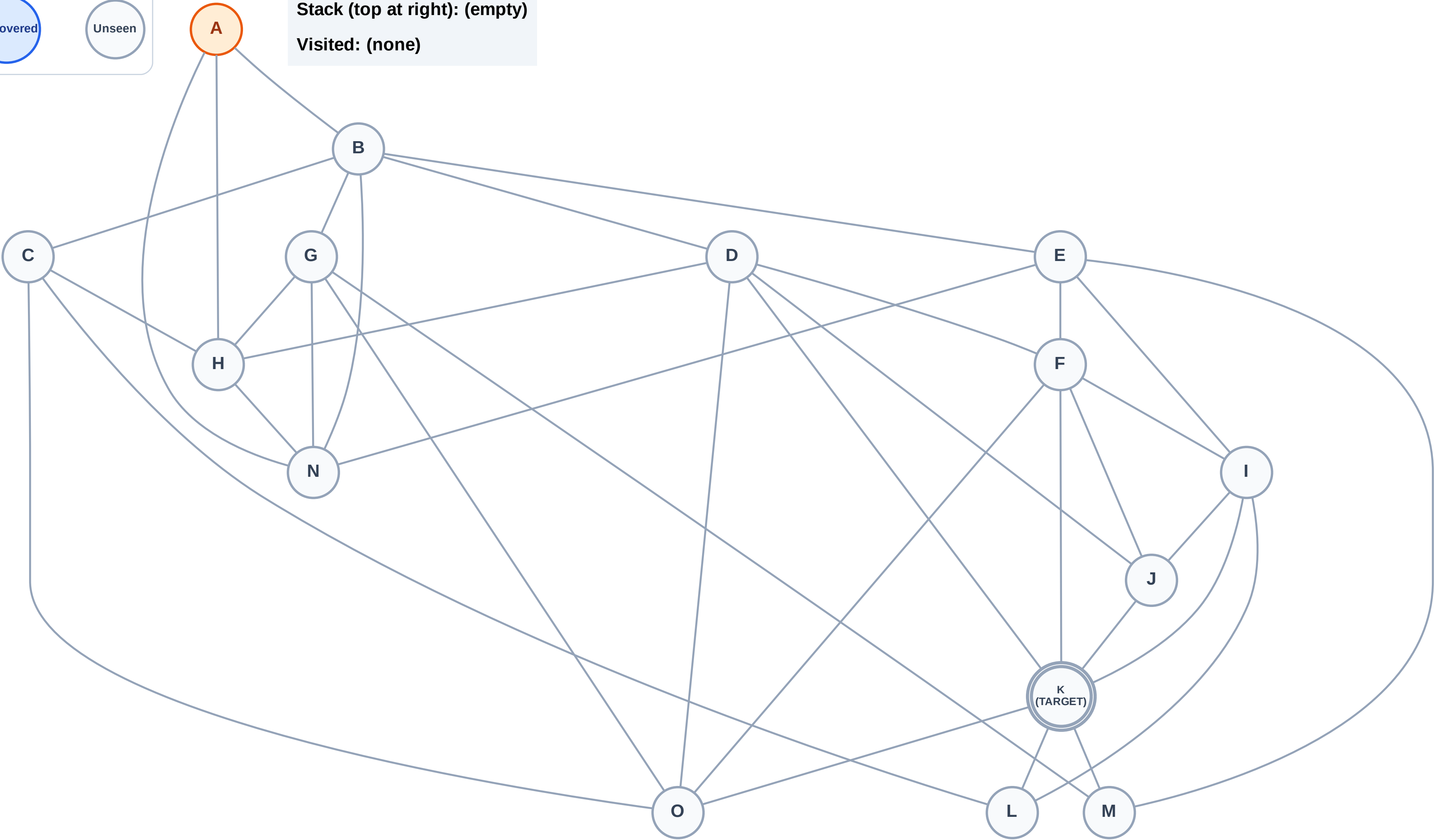
Complete

Processing

Discovered

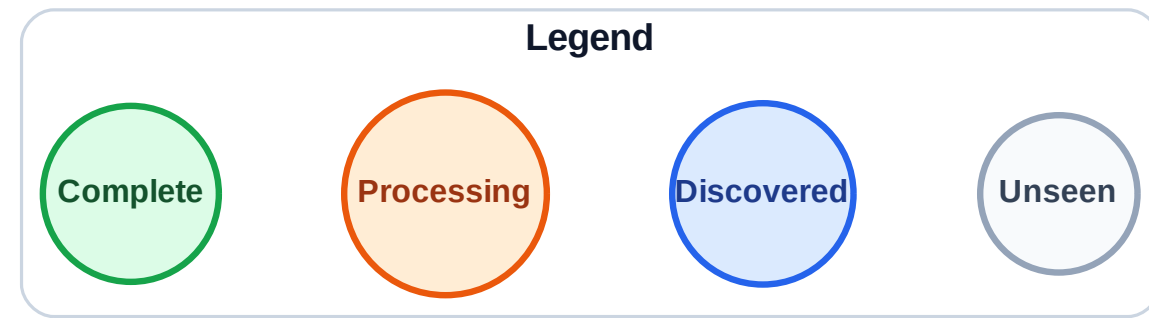
Unseen

Stack (top at right): (empty)
Visited: (none)

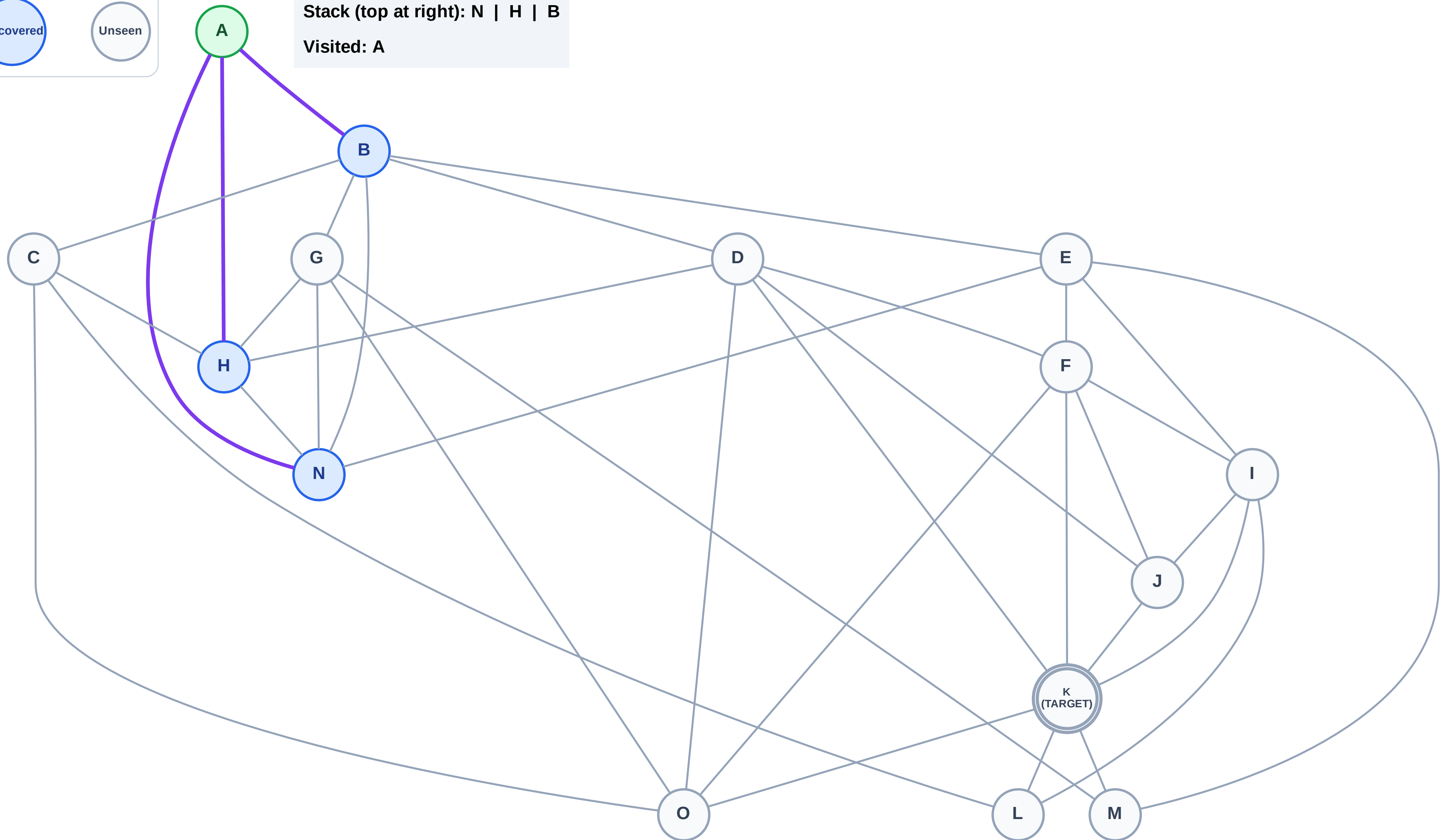


DFS Traversal

Step 3: Discover N, H, B from A



Stack (top at right): N | H | B
Visited: A



DFS Traversal

Step 4: Process node B

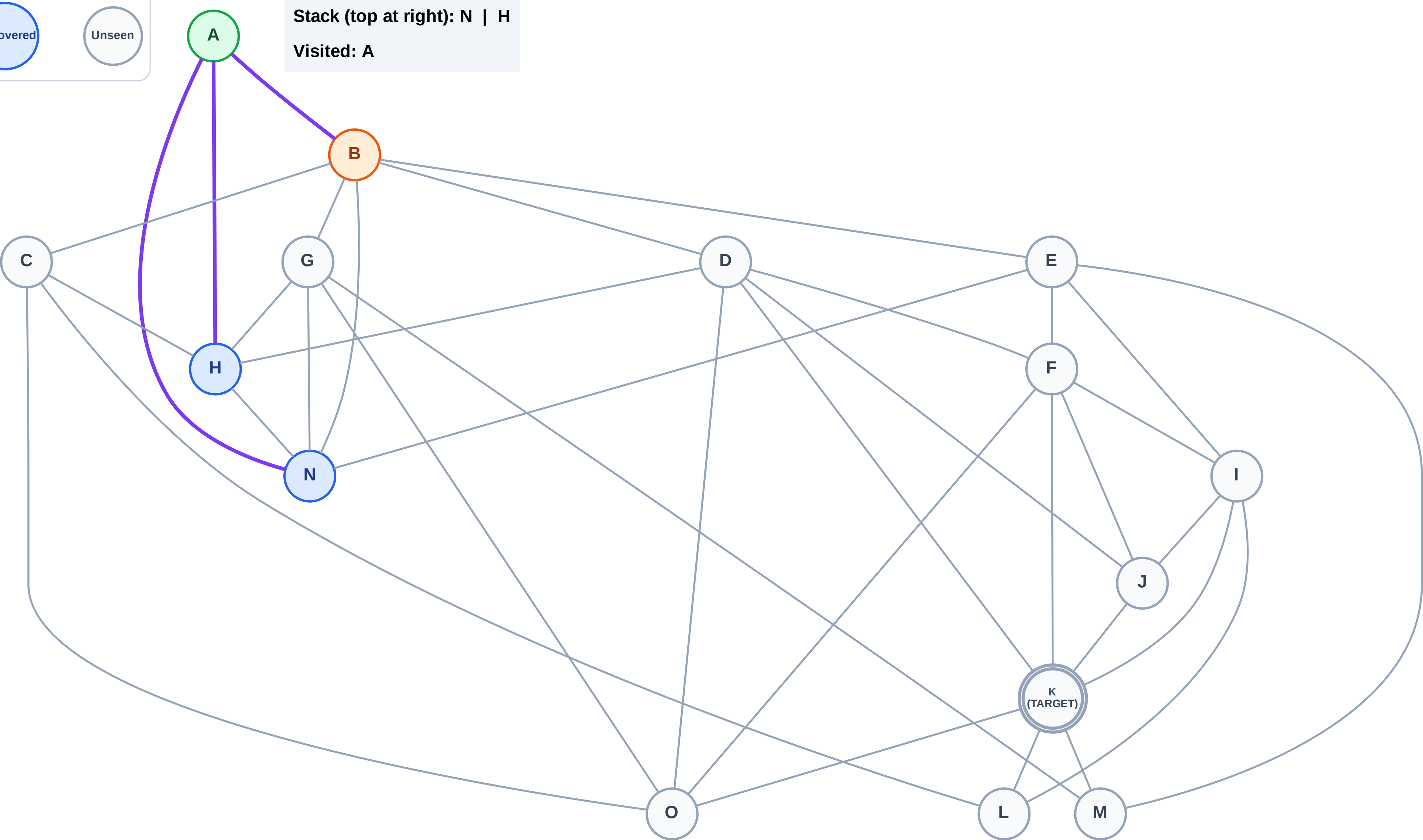
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 5: Discover G, E, D, C from B

Legend

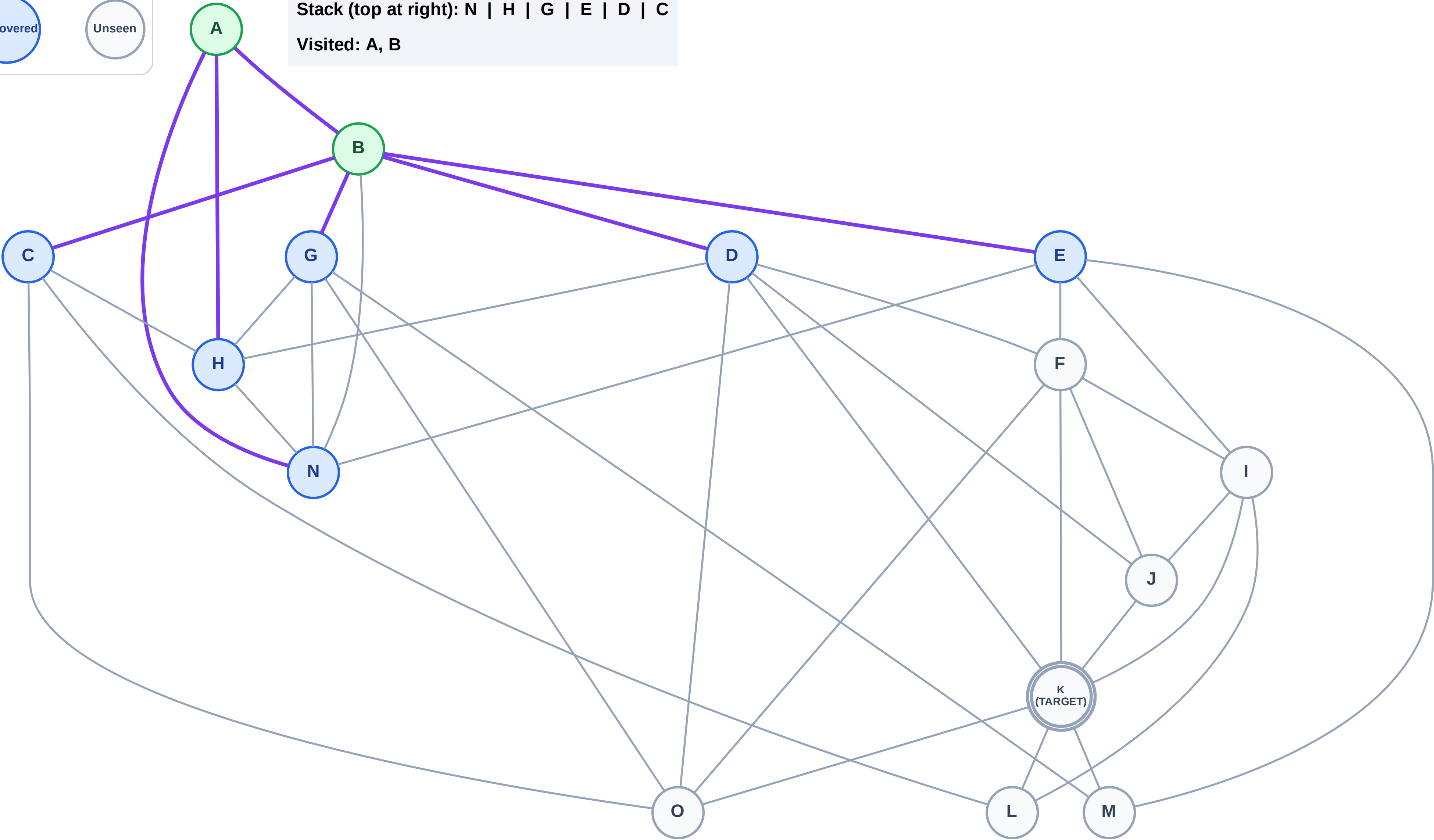
Complete

Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | C
Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

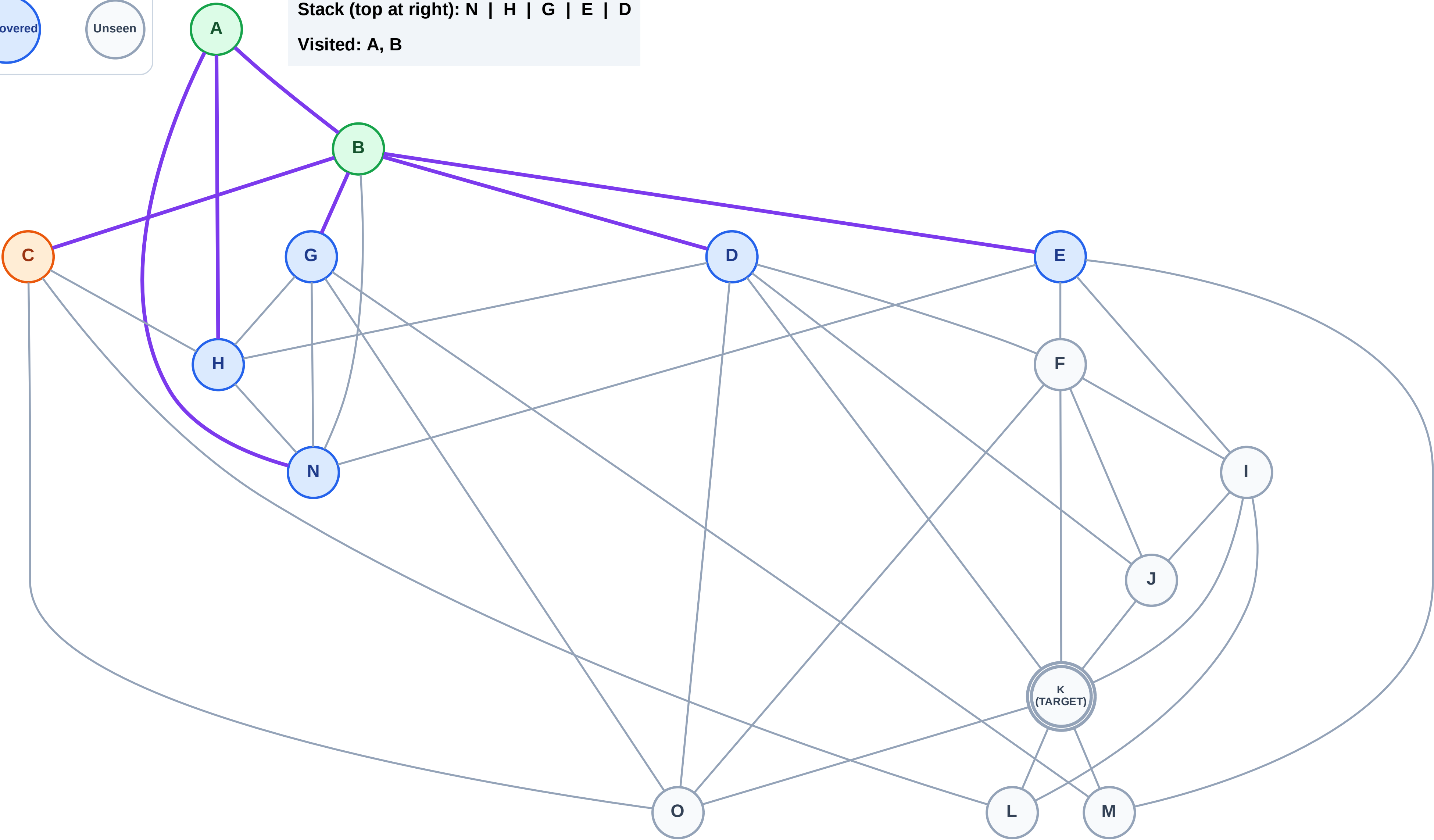
Complete

Processing

Discovered

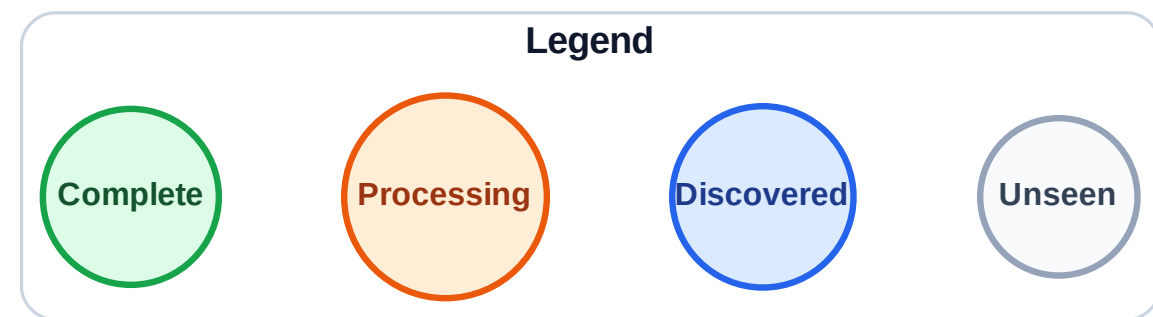
Unseen

Stack (top at right): N | H | G | E | D
Visited: A, B



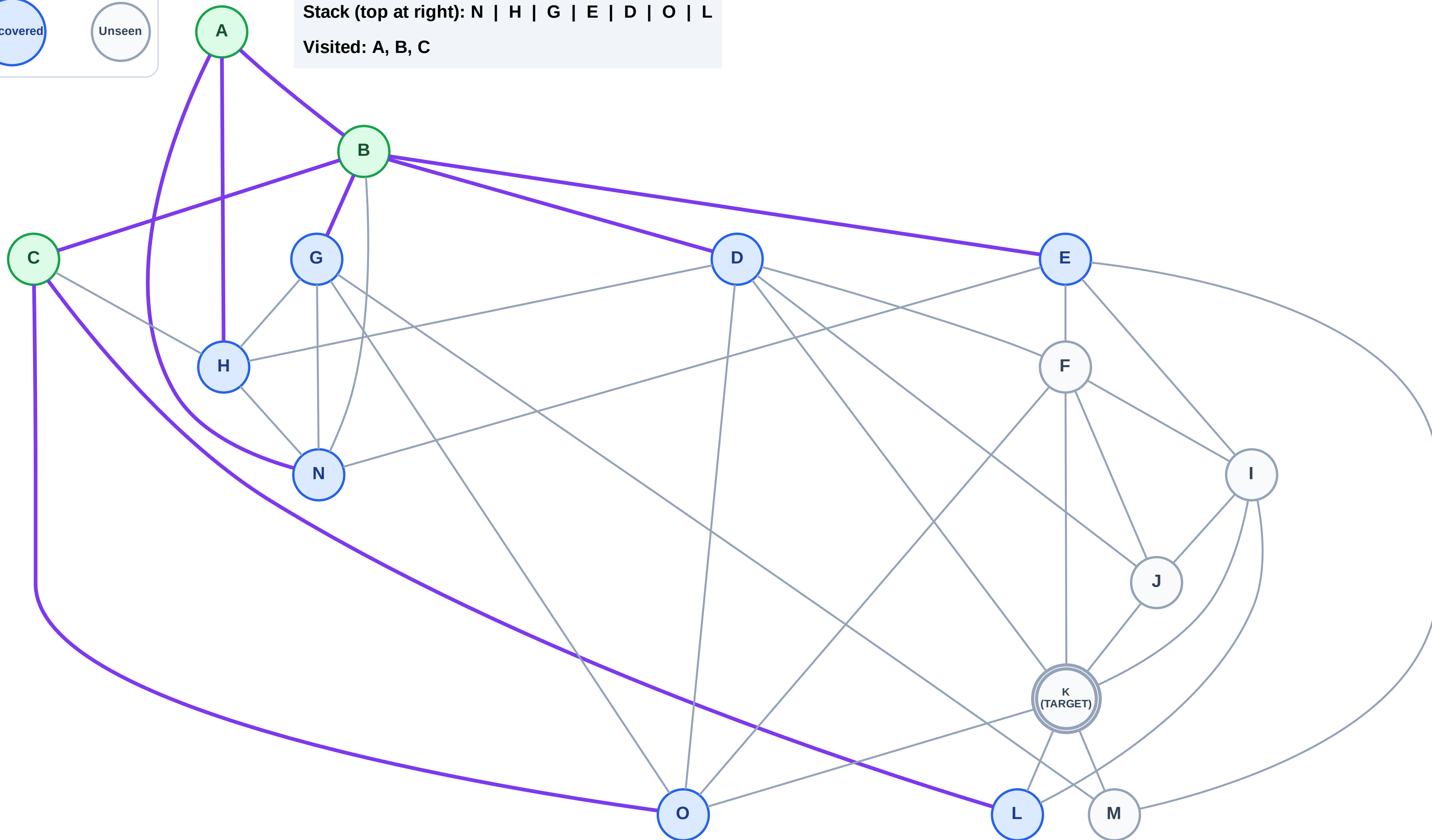
DFS Traversal

Step 7: Discover O, L from C



Stack (top at right): N | H | G | E | D | O | L

Visited: A, B, C



DFS Traversal

Step 8: Process node L

Legend

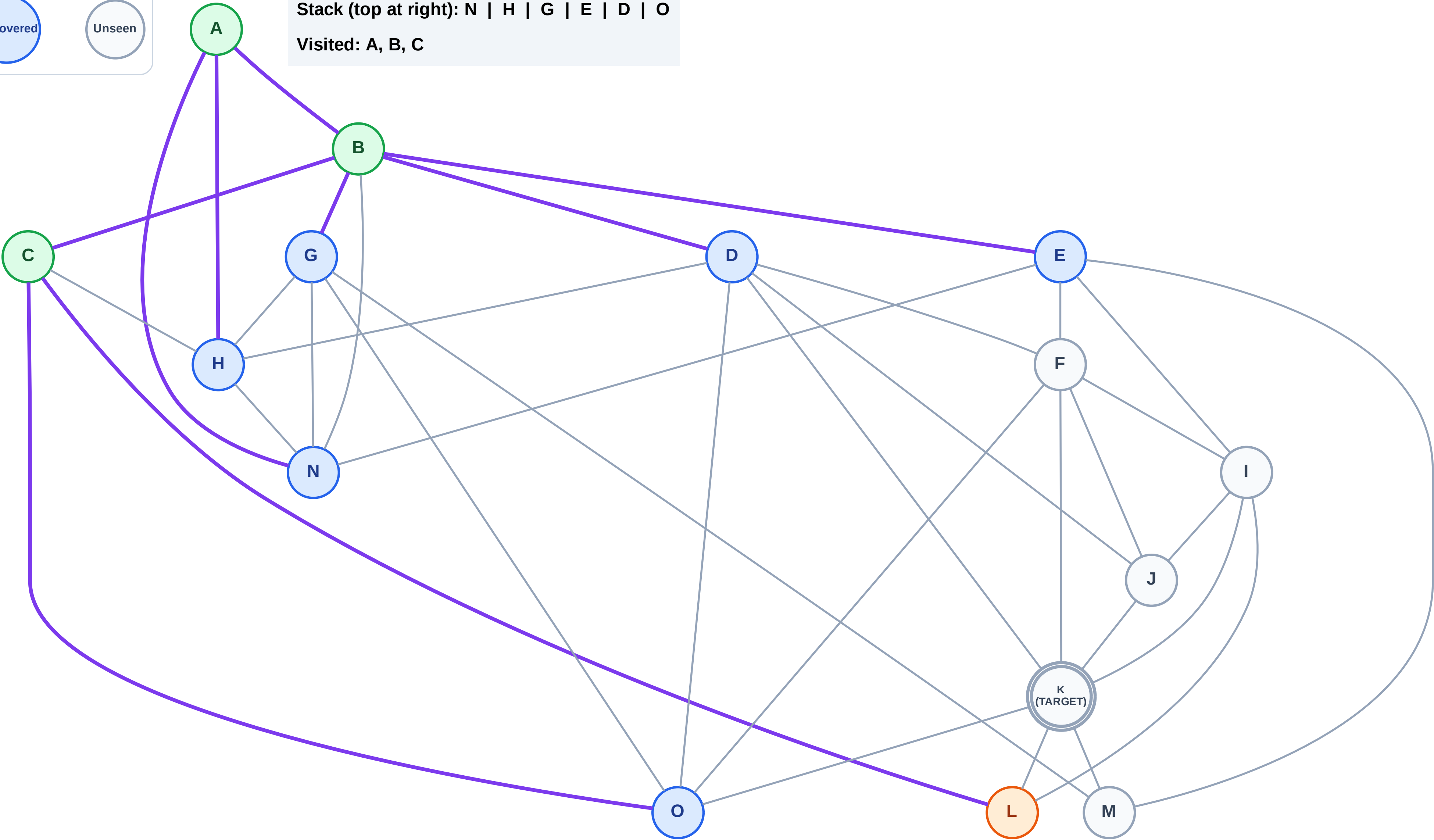
Complete

Processing

Discovered

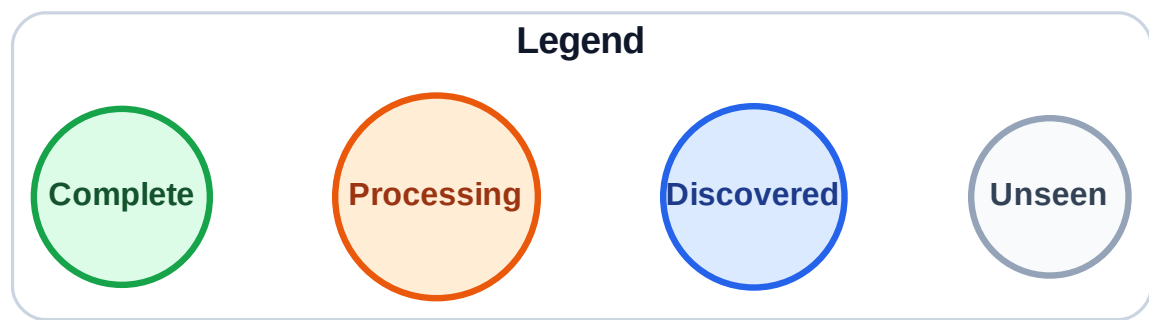
Unseen

Stack (top at right): N | H | G | E | D | O
Visited: A, B, C



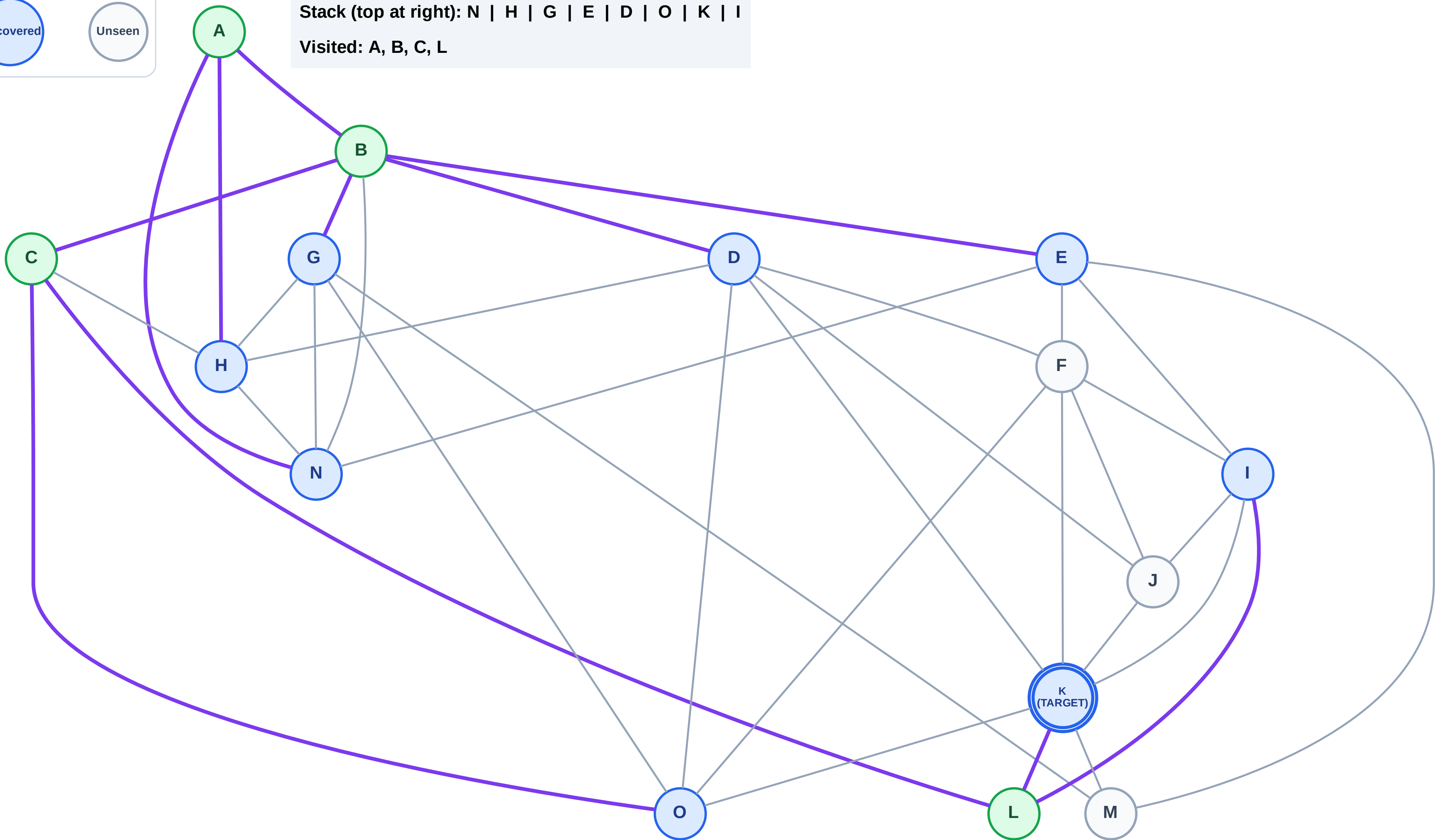
DFS Traversal

Step 9: Discover K, I from L



Stack (top at right): N | H | G | E | D | O | K | I

Visited: A, B, C, L



DFS Traversal

Step 10: Process node I

Legend

Complete

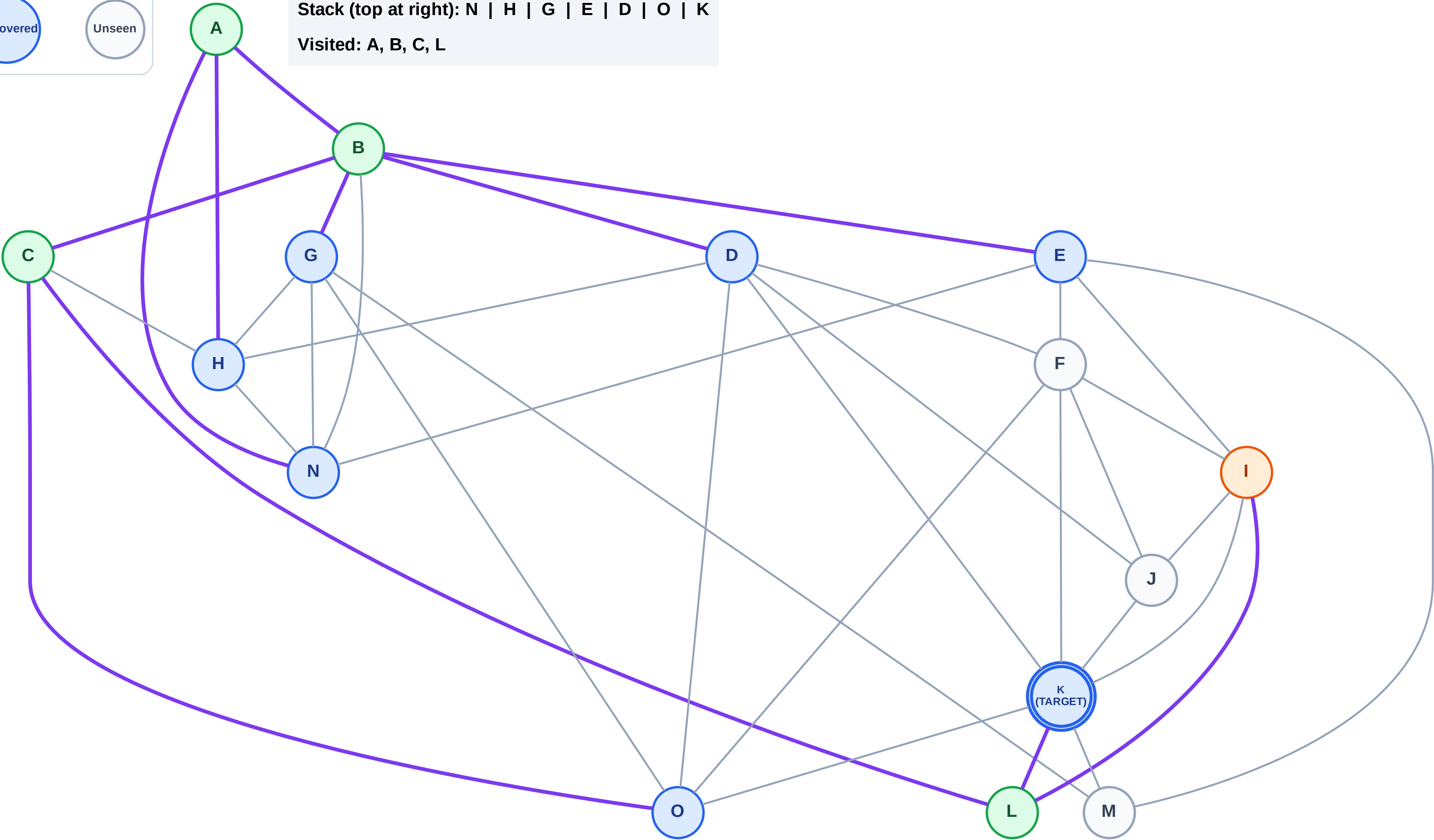
Processing

Discovered

Unseen

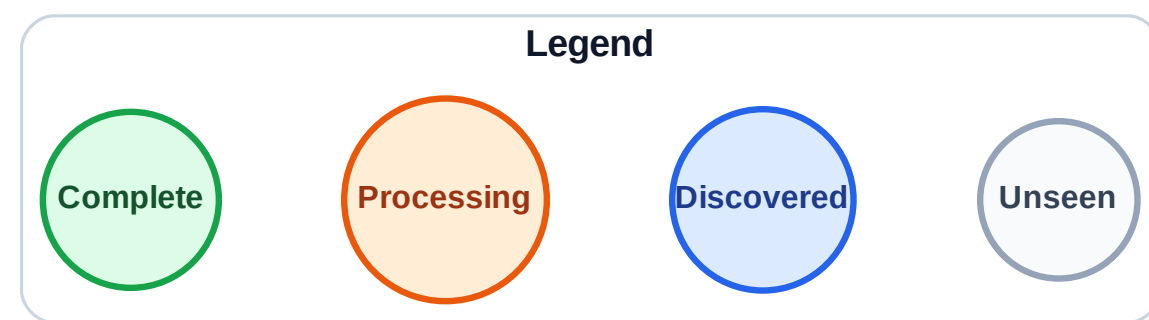
Stack (top at right): N | H | G | E | D | O | K

Visited: A, B, C, L



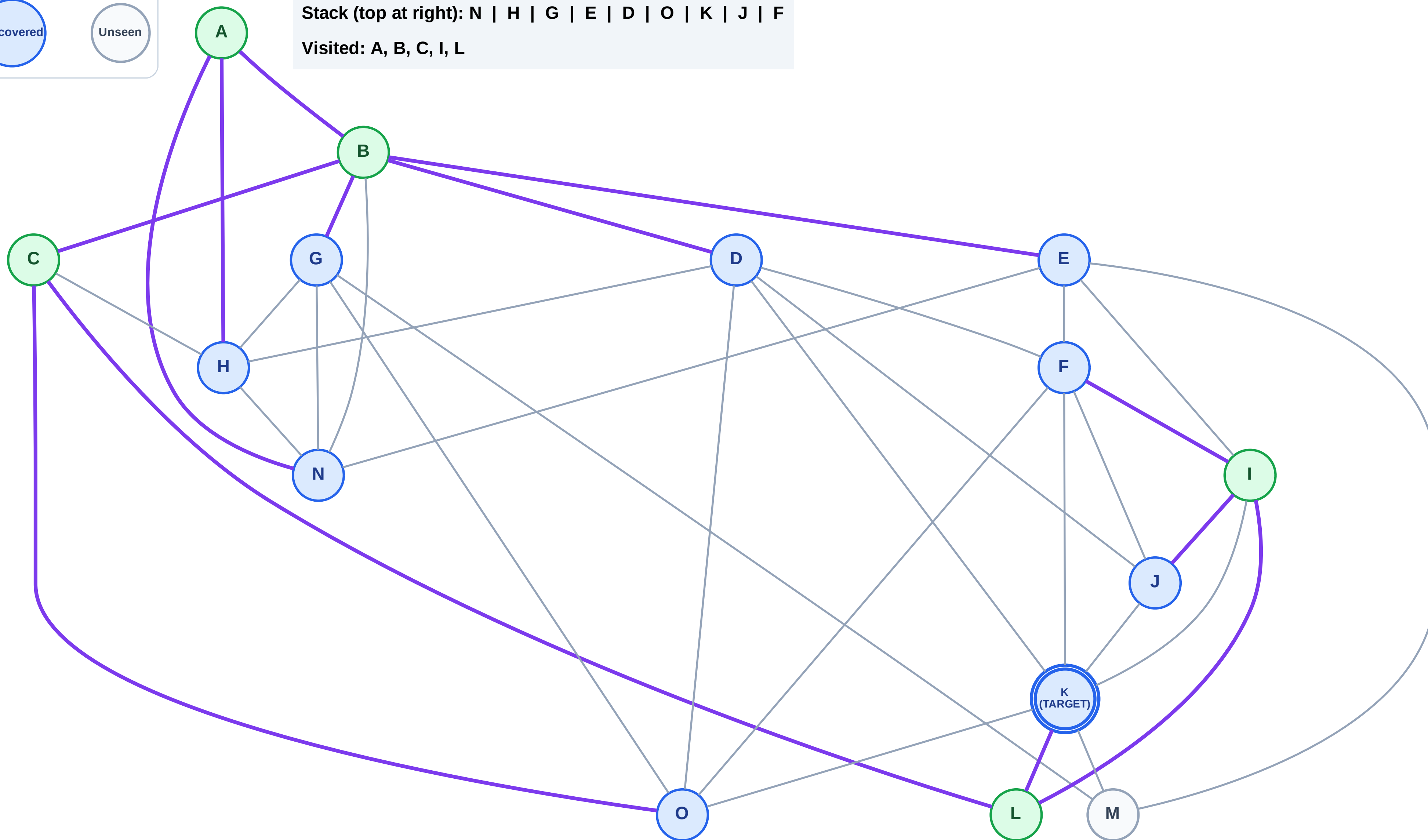
DFS Traversal

Step 11: Discover J, F from I



Stack (top at right): N | H | G | E | D | O | K | J | F

Visited: A, B, C, I, L



DFS Traversal

Step 12: Process node F

Legend

Complete

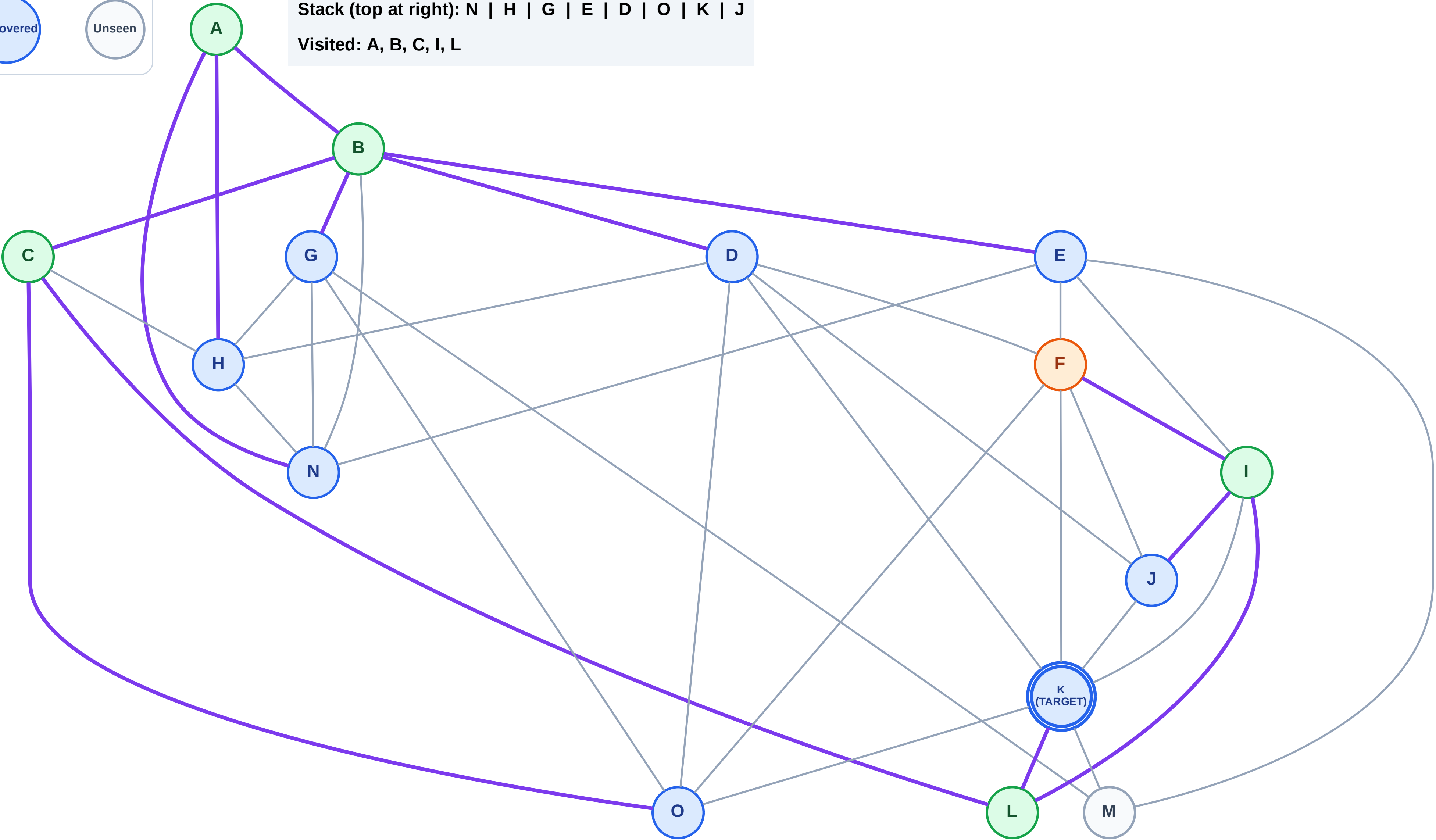
Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | O | K | J

Visited: A, B, C, I, L



DFS Traversal

Step 13: No new neighbors from F

Legend

Complete

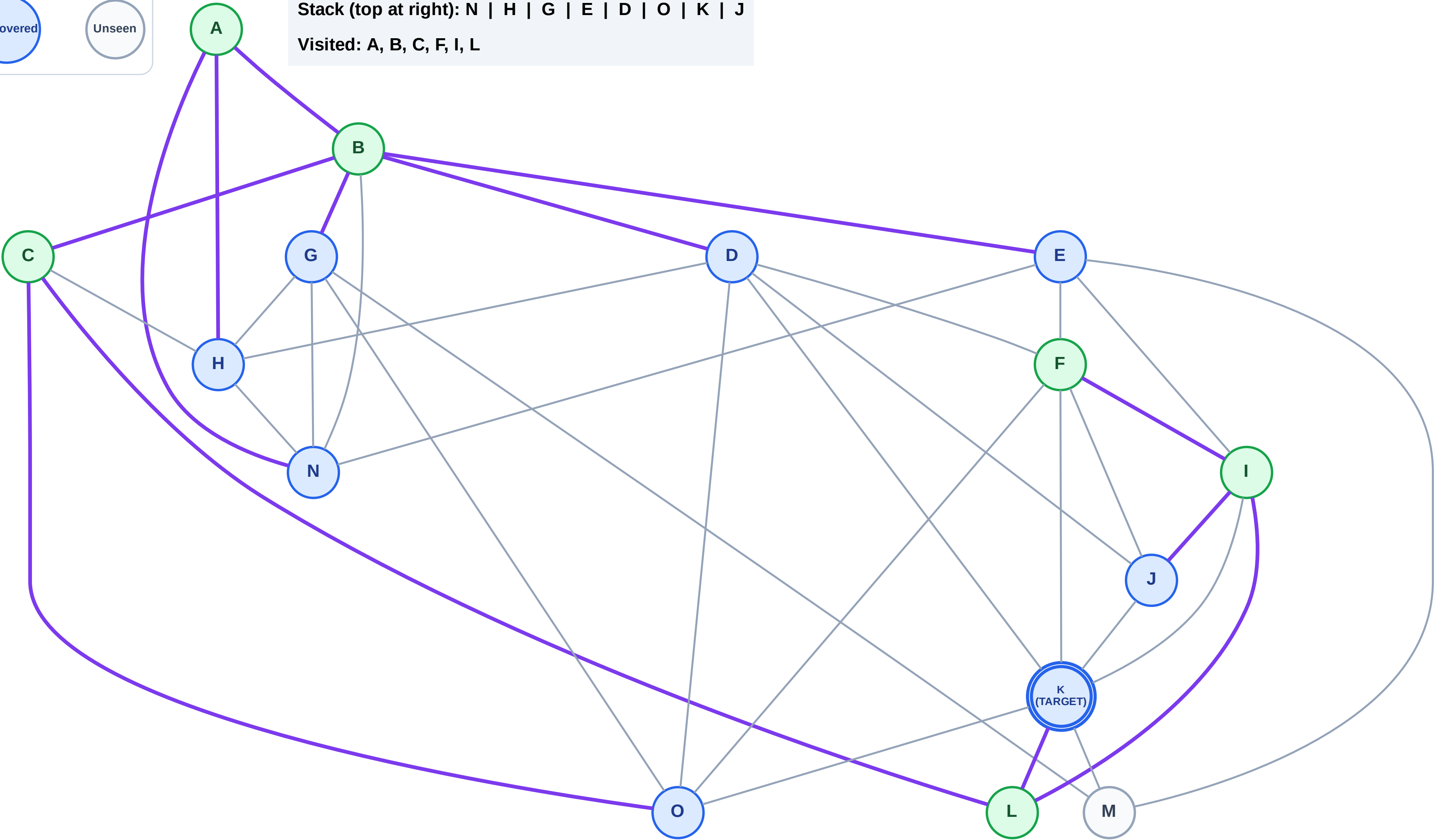
Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | O | K | J

Visited: A, B, C, F, I, L



DFS Traversal

Step 14: Process node J

Legend

Complete

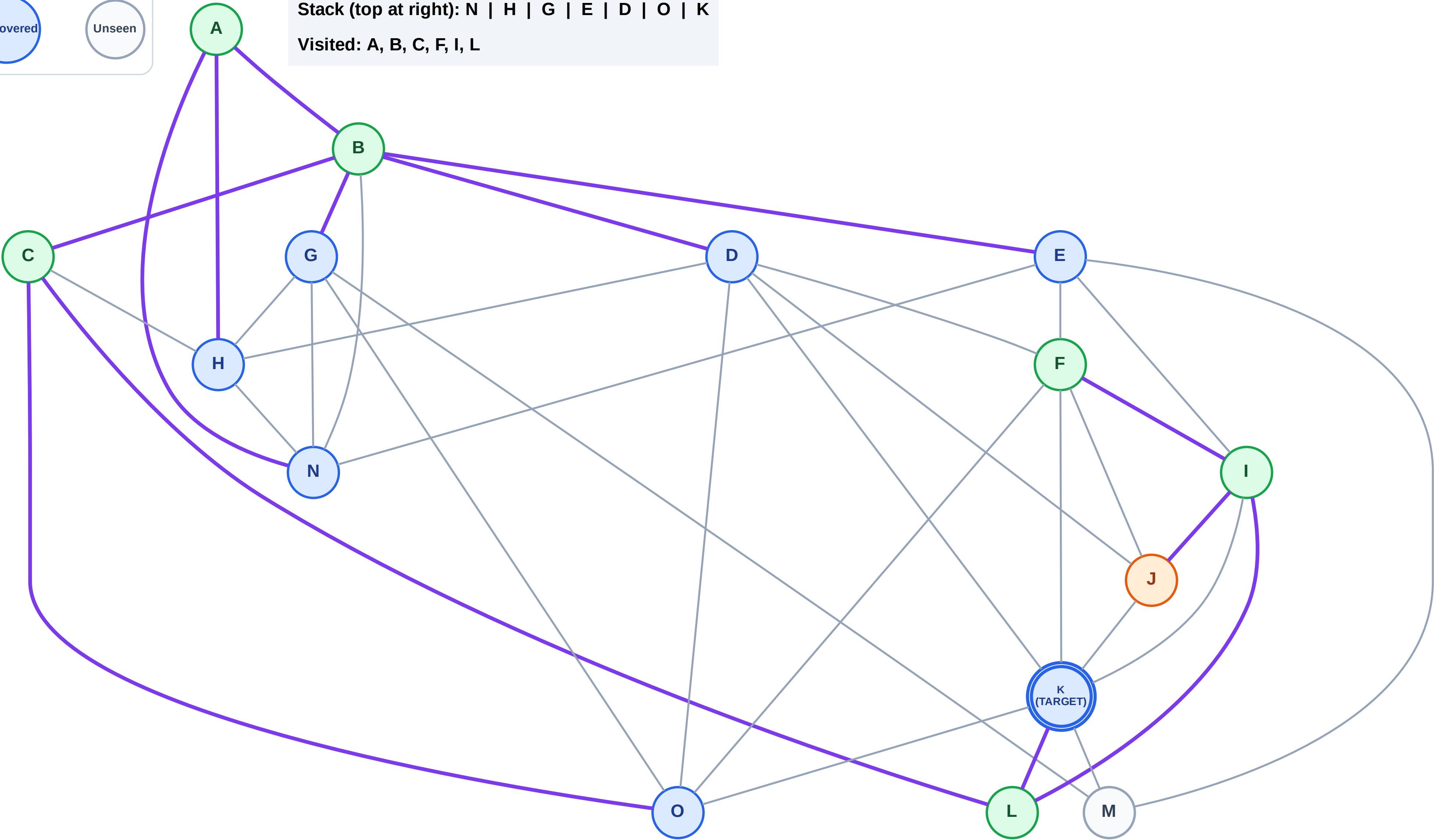
Processing

Discovered

Unseen

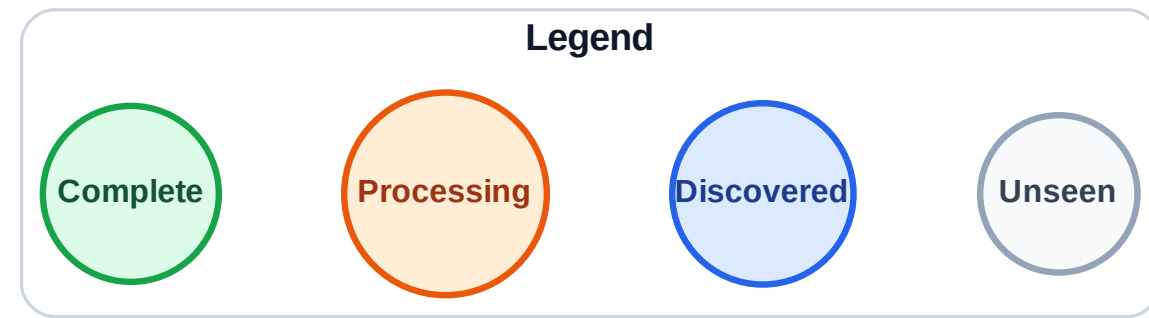
Stack (top at right): N | H | G | E | D | O | K

Visited: A, B, C, F, I, L

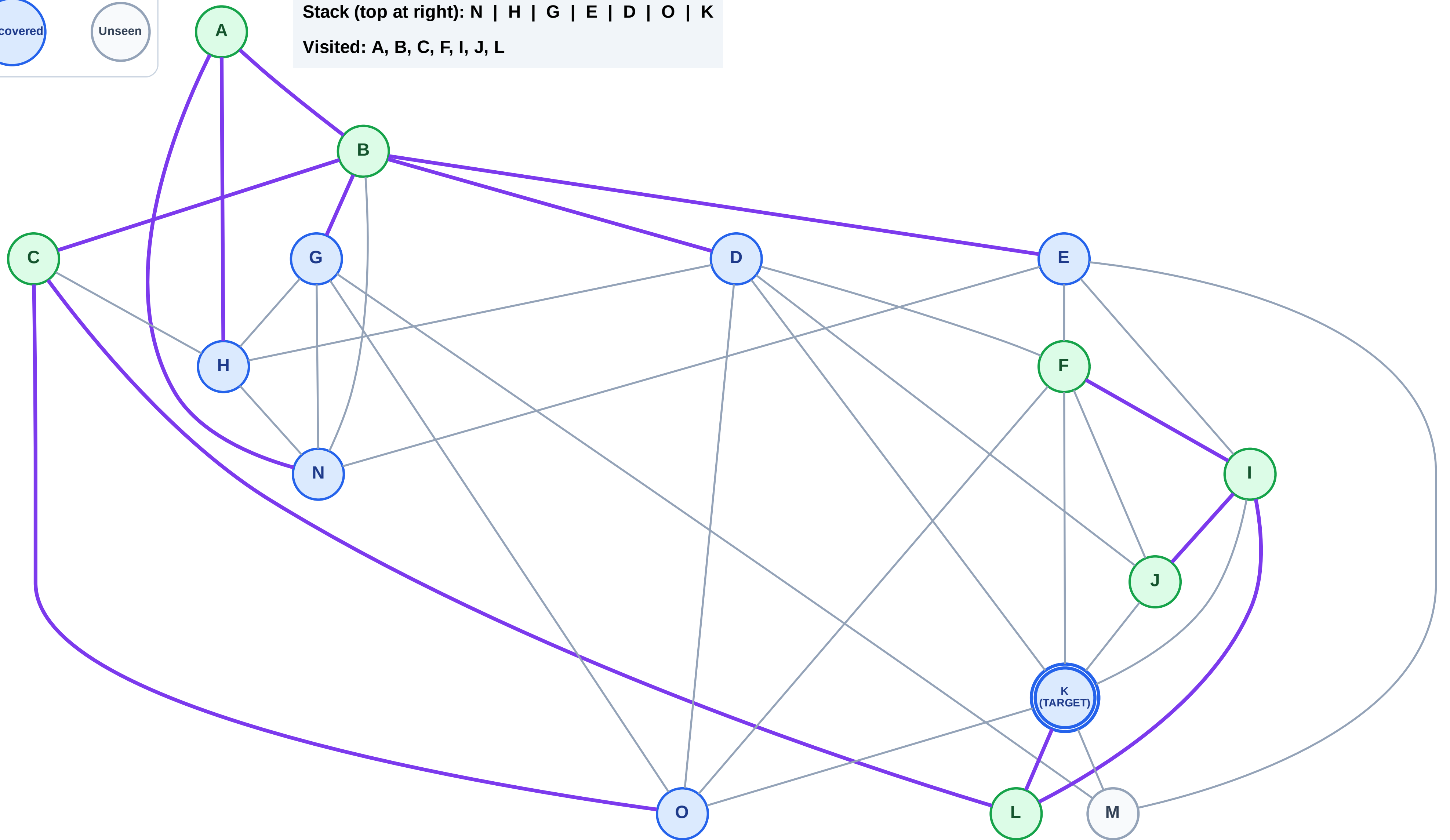


DFS Traversal

Step 15: No new neighbors from J



Stack (top at right): N | H | G | E | D | O | K
Visited: A, B, C, F, I, J, L



DFS Traversal

Step 16: Process node K

Legend

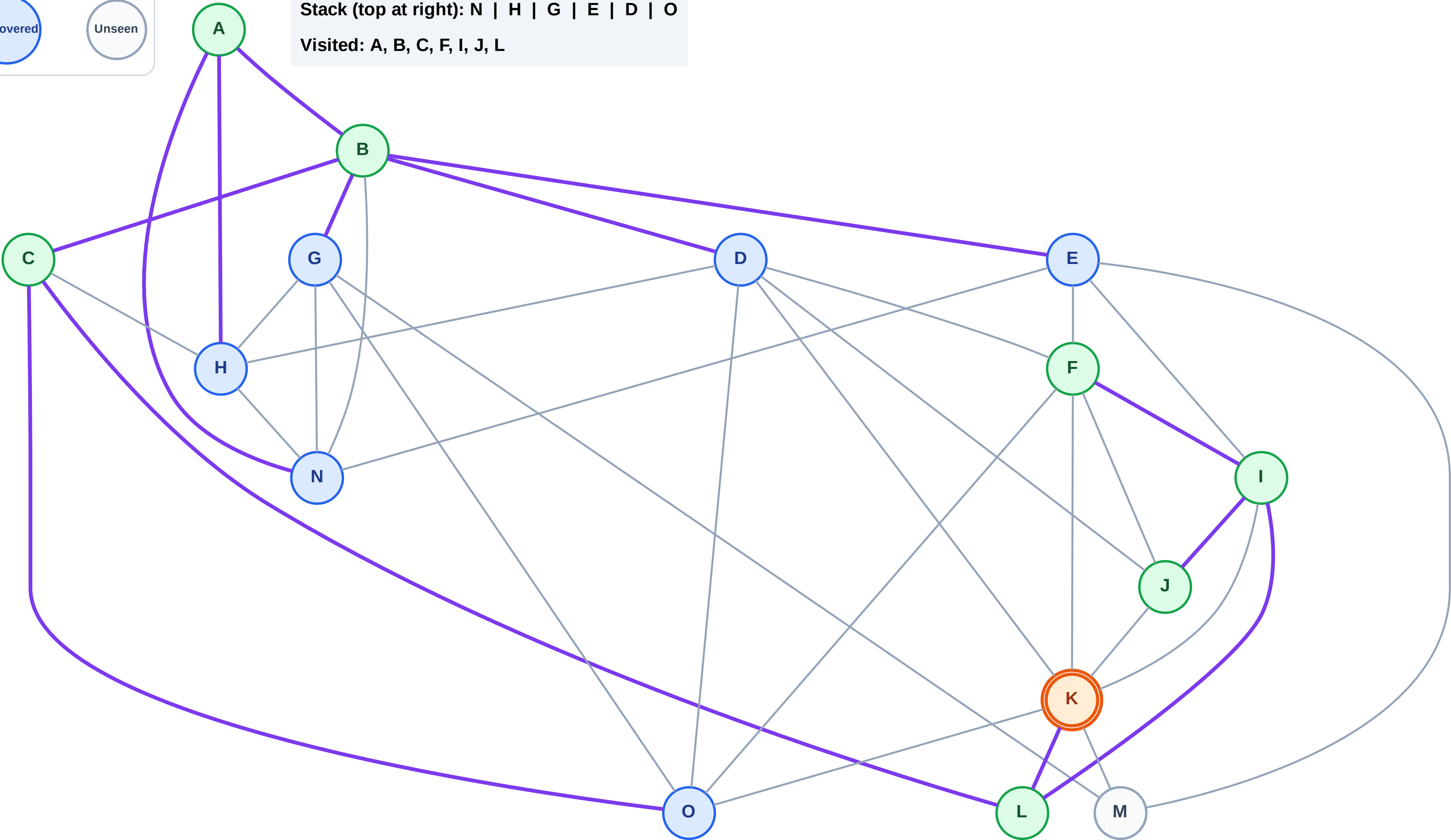
Complete

Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | O
Visited: A, B, C, F, I, J, L



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → C → L → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | H | G | E | D | O
Visited: A, B, C, F, I, J, K, L

